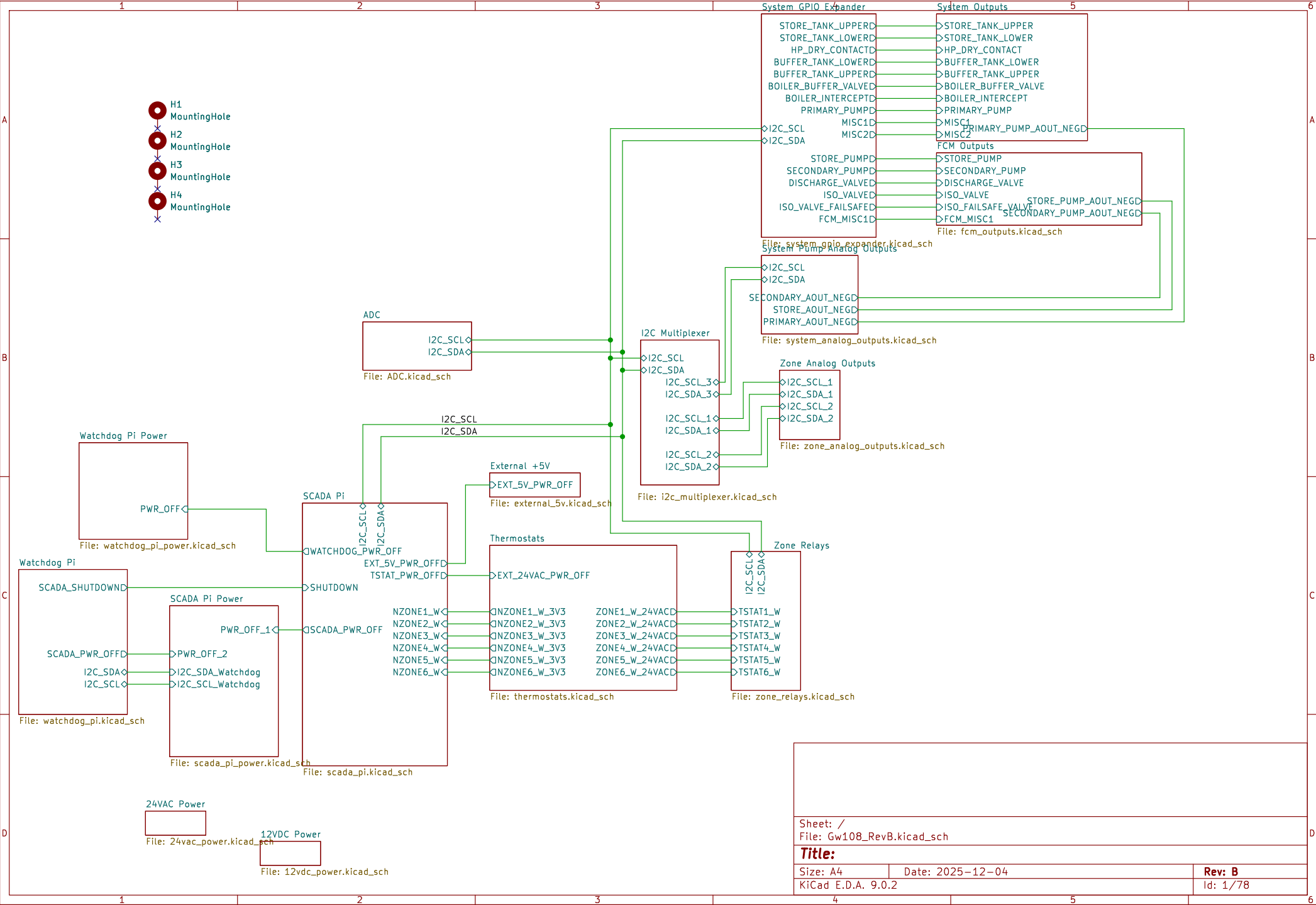
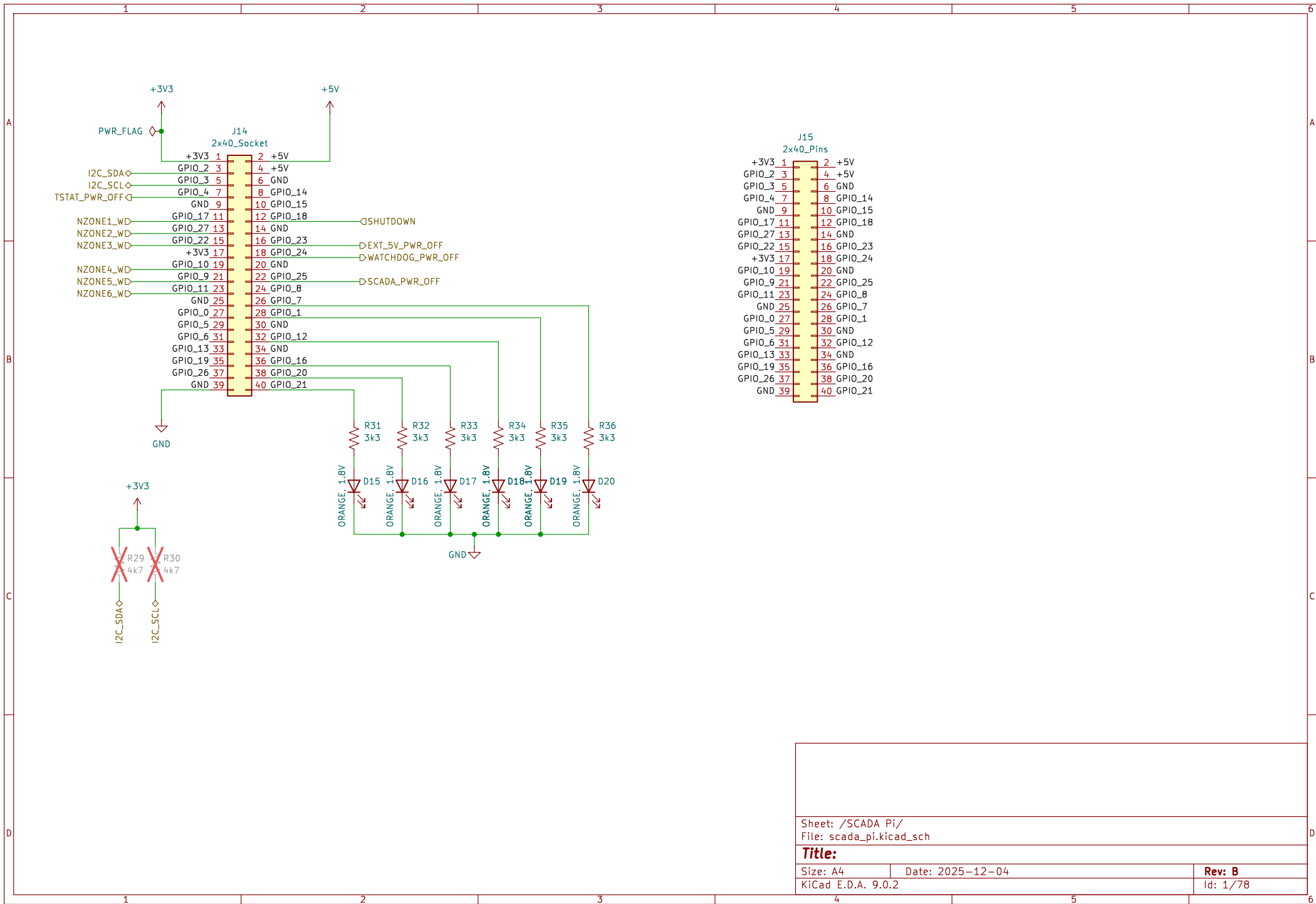






Sheet: /SCADA Pi/
File: scada_pi.kicad_sch

Title:

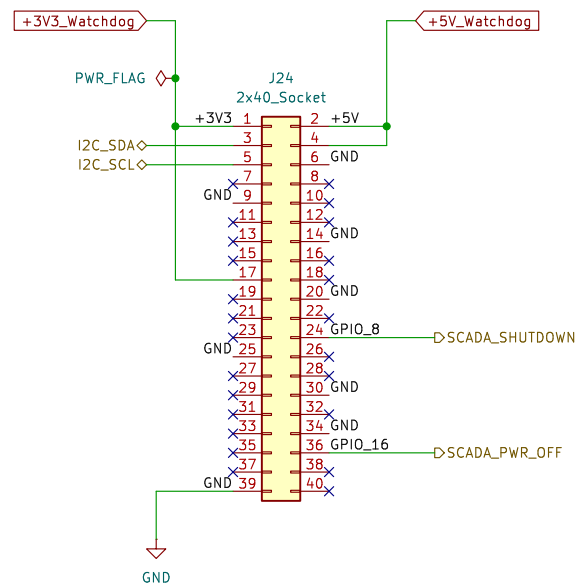
Size: A4

Date: 2025-12-04

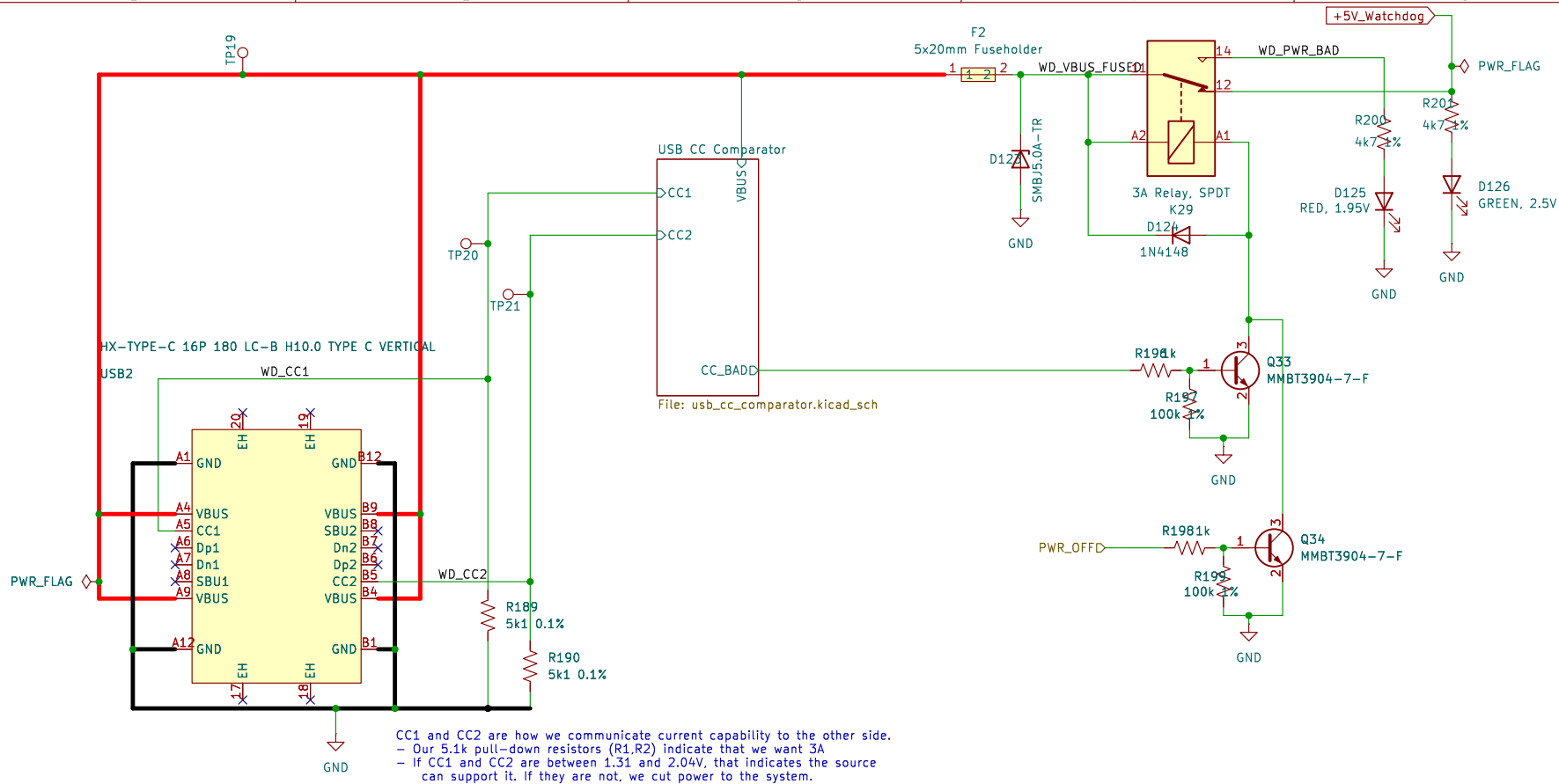
KiCad E.D.A. 9.0.2

Rev: B

Id: 1/78



Sheet: /Watchdog PI/ File: watchdog_pi.kicad_sch		
Title:		
Size: A4	Date: 2025-12-04	Rev: B
KiCad E.D.A. 9.0.2	Id: 3/78	



CC1 and CC2 are how we communicate current capability to the other side.
 - Our 5.1k pull-down resistors (R1,R2) indicate that we want 3A
 - If CC1 and CC2 are between 1.31 and 2.04V, that indicates the source can support it. If they are not, we cut power to the system.

Sheet: /Watchdog PI Power/
 File: watchdog_pi_power.kicad_sch

Title:

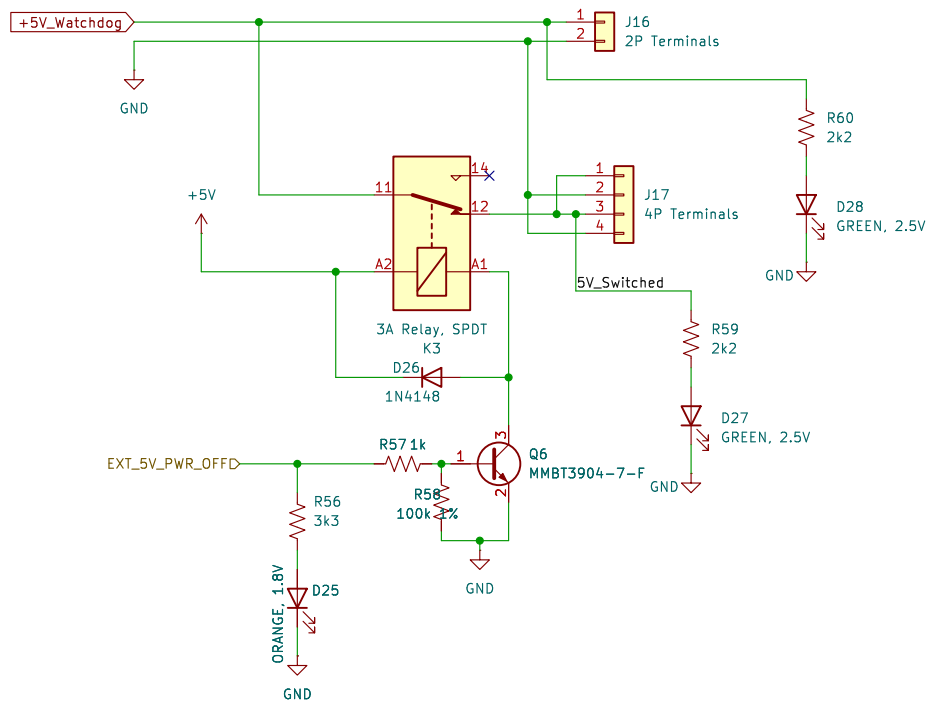
Size: A4

Date: 2025-12-04

KiCad E.D.A. 9.0.2

Rev: B

Id: 4/78



Sheet: /External +5V/
File: externalL5v.kicad_sch

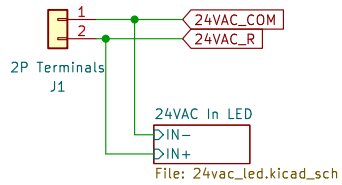
Title:

Size: A4 Date: 2025-12-04

KiCad E.D.A. 9.0.2

Rev: B

Id: 5/78



Sheet: /24VAC Power/
File: 24vac_power.kicad_sch

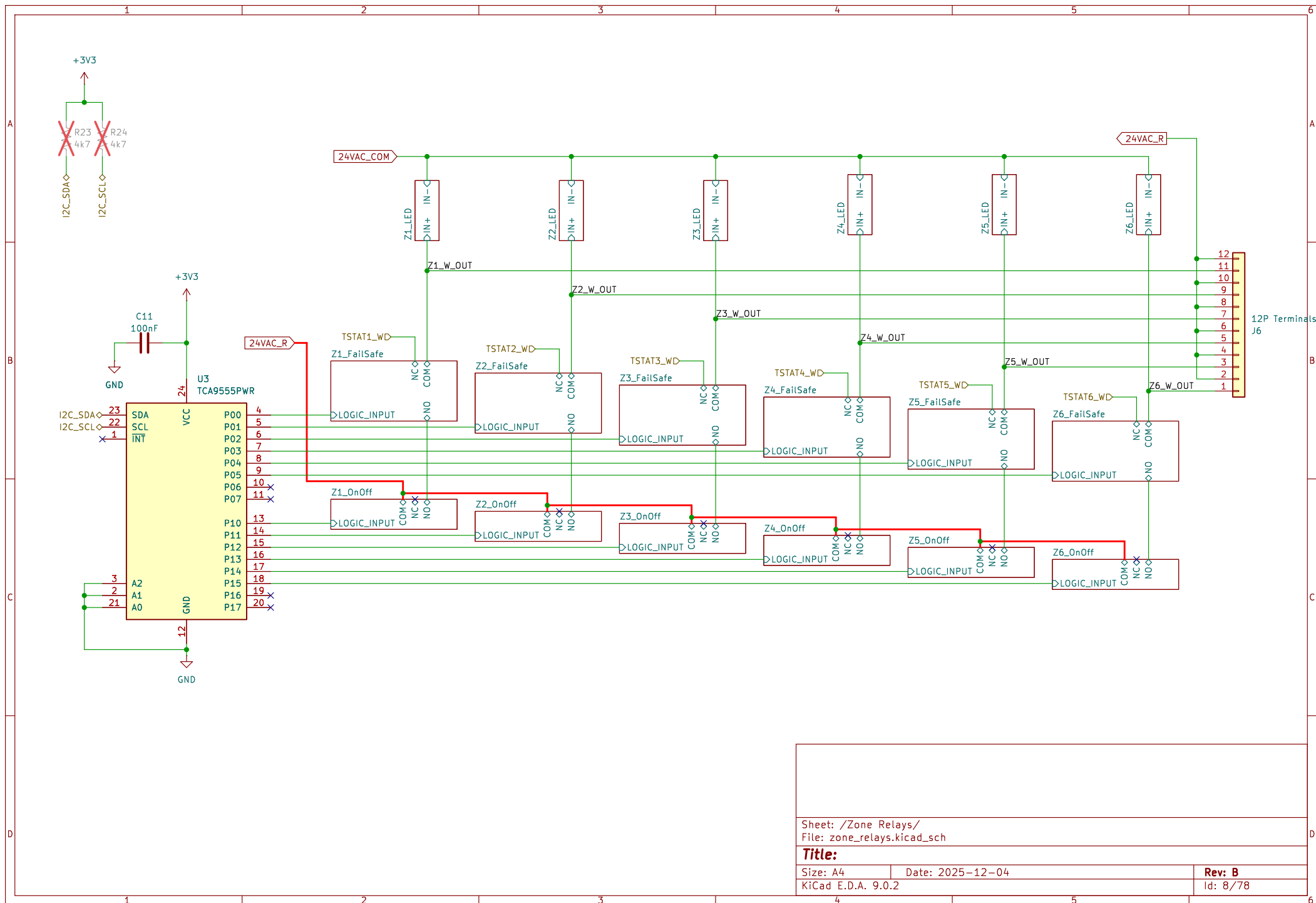
Title:

Size: A4 Date: 2025-12-04

KiCad E.D.A. 9.0.2

Rev: B

Id: 6/78



Sheet: /Zone Relays/
File: zone_relays.kicad_sch

Title:

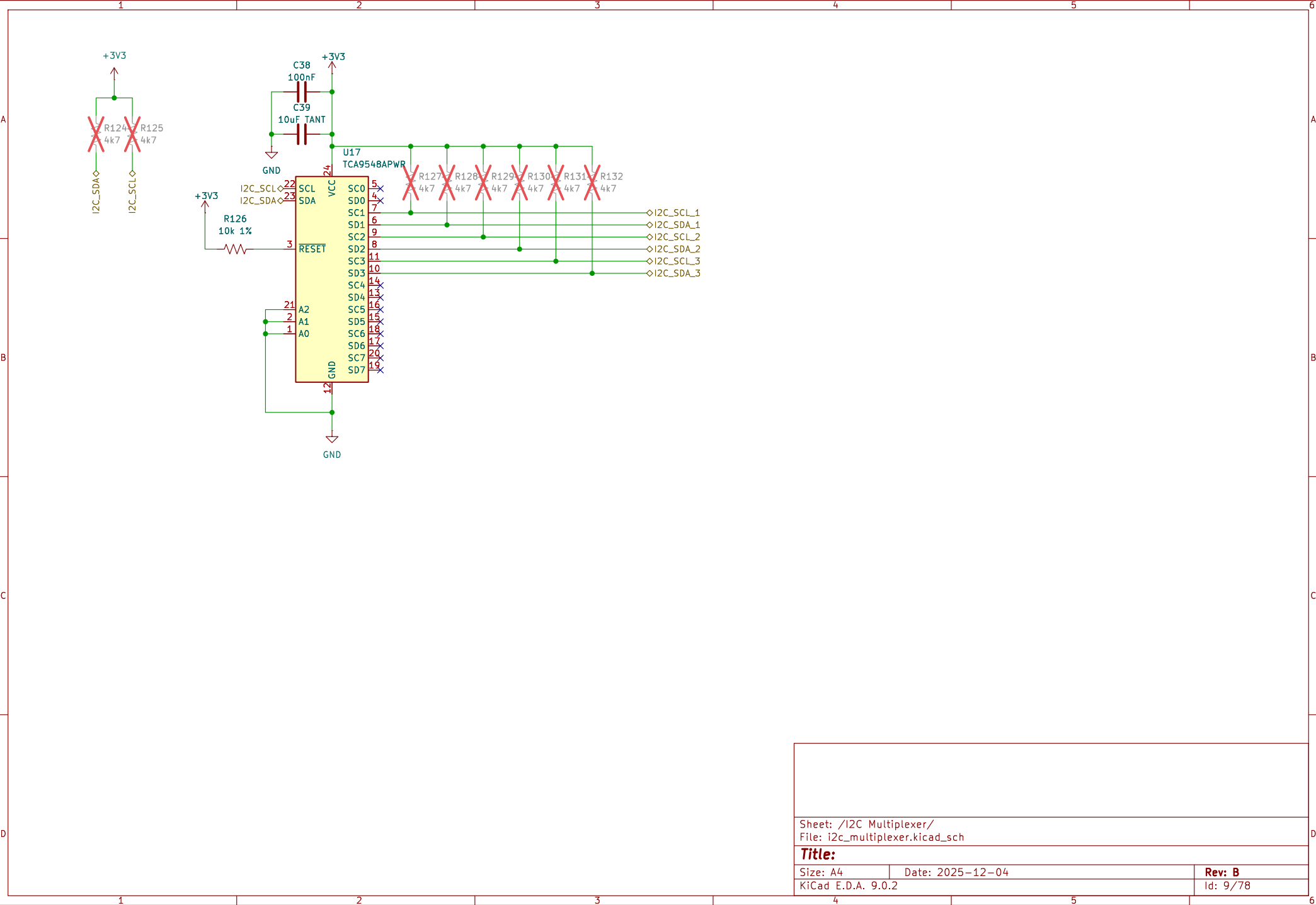
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 8/78



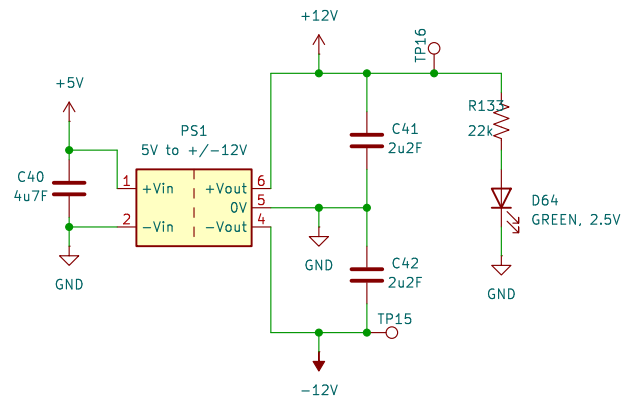
Sheet: /I2C Multiplexer/
File: i2c_mux.kicad_sch

Title:

Size: A4
KiCad E.D.A. 9.0.2

Date: 2025-12-04

Rev: B
Id: 9/78



Sheet: /12VDC Power/
File: 12vdc_power.kicad_sch

Title:

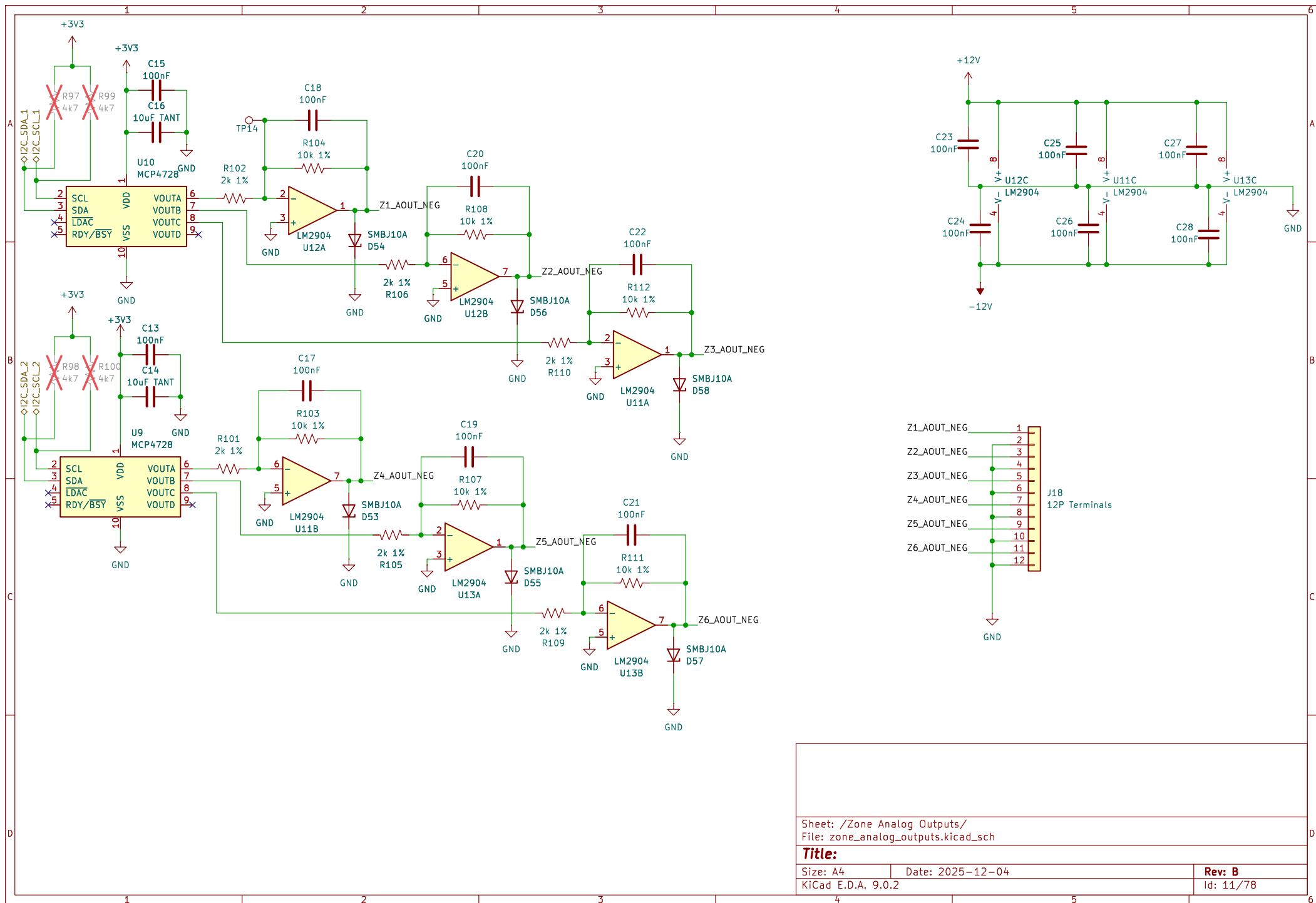
Size: A4

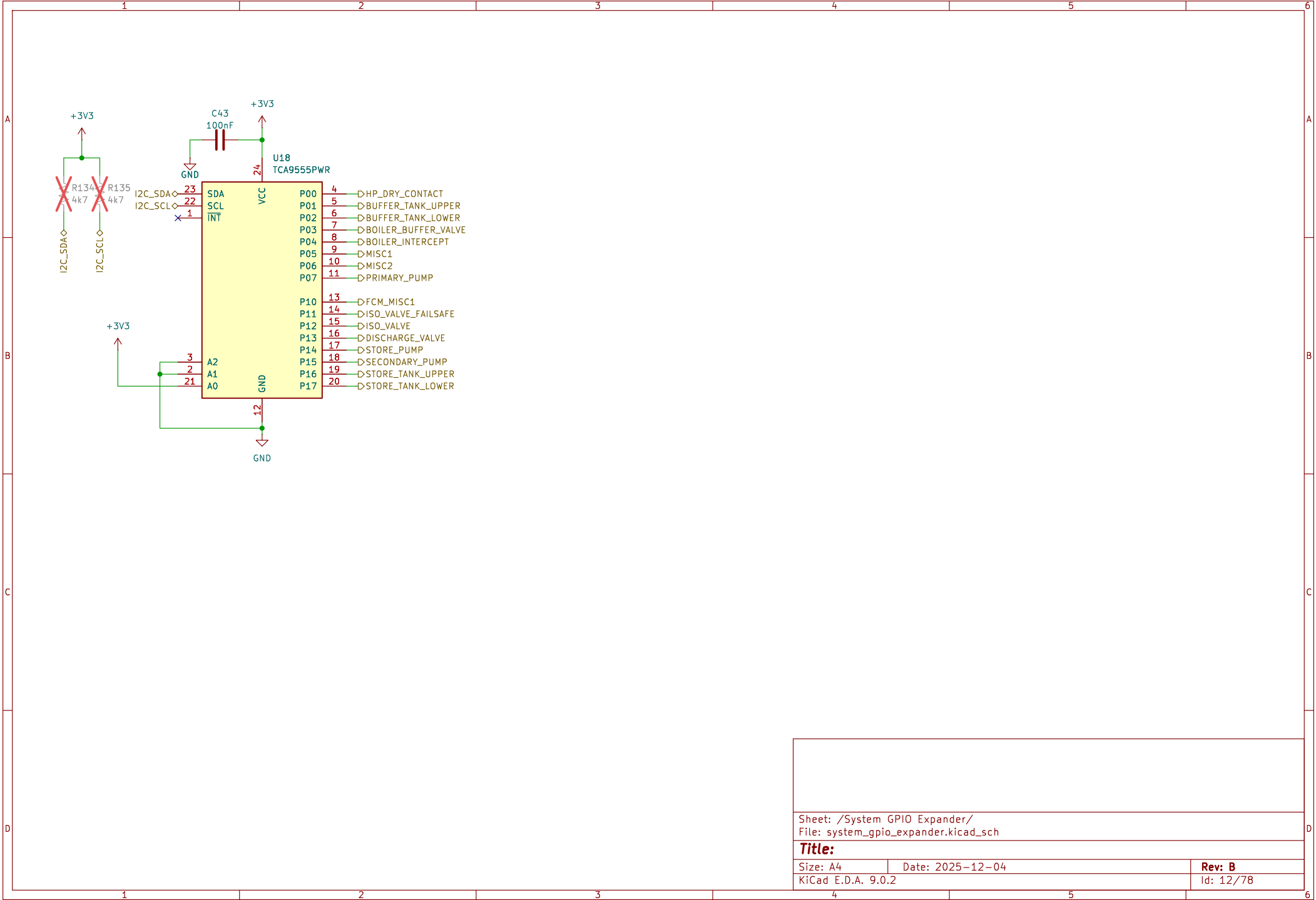
Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 10/78





Sheet: /System GPIO Expander/
File: system_gpio_expander.kicad_sch

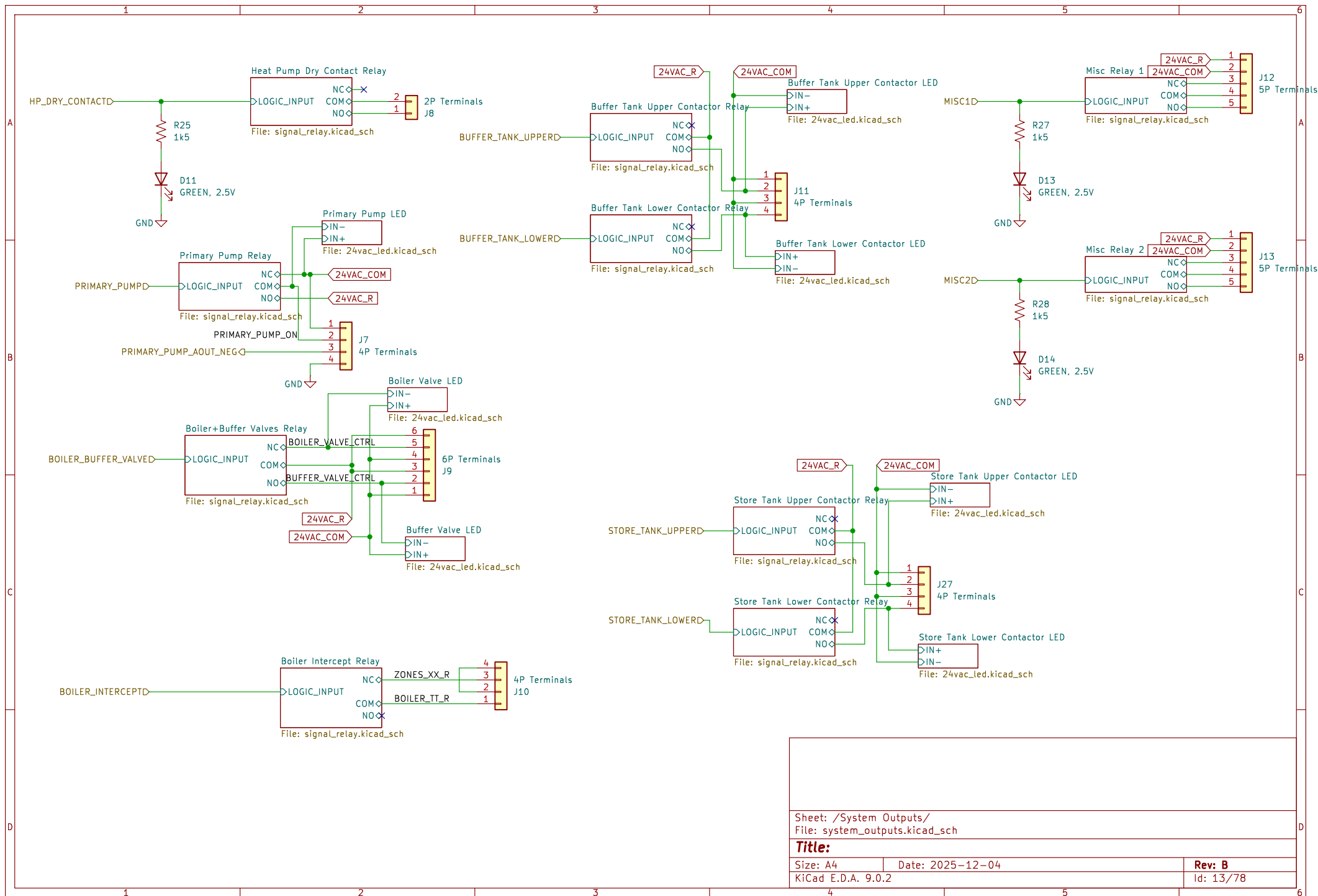
Title:

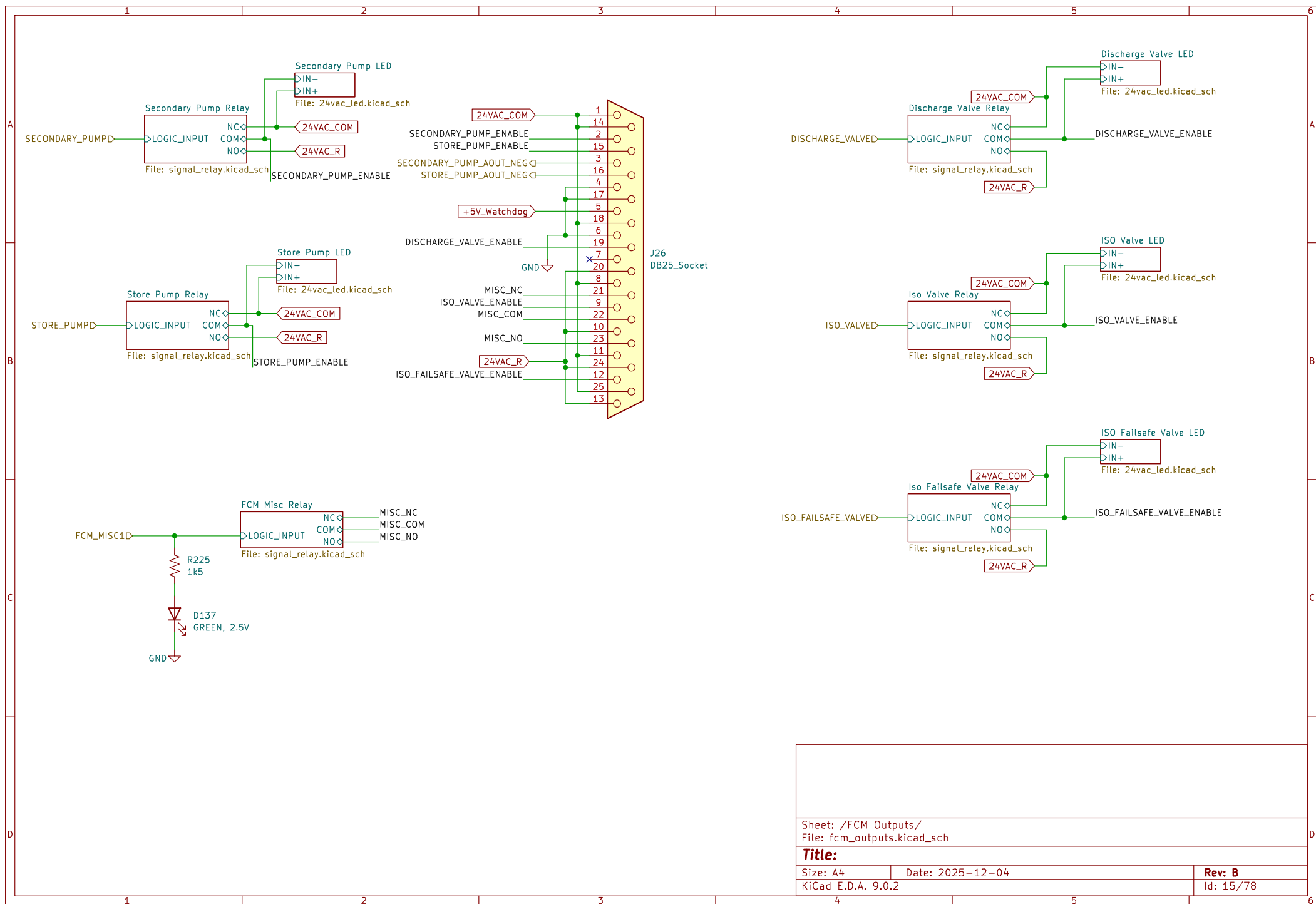
Size: A4
KiCad E.D.A. 9.0.2

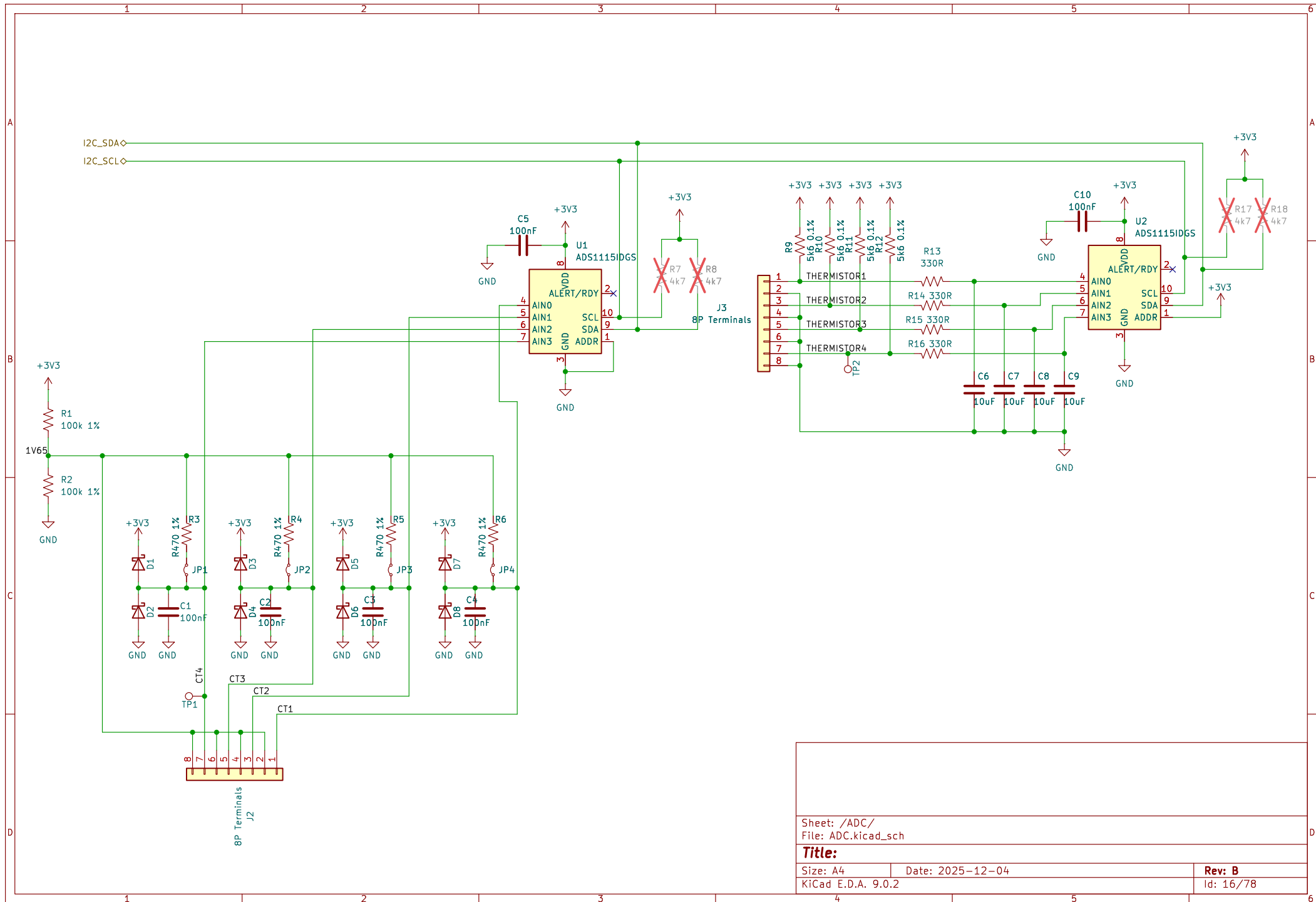
Date: 2025-12-04

Rev: B

Id: 12/78







Sheet: /ADC/
File: ADC.kicad_sch

Title:

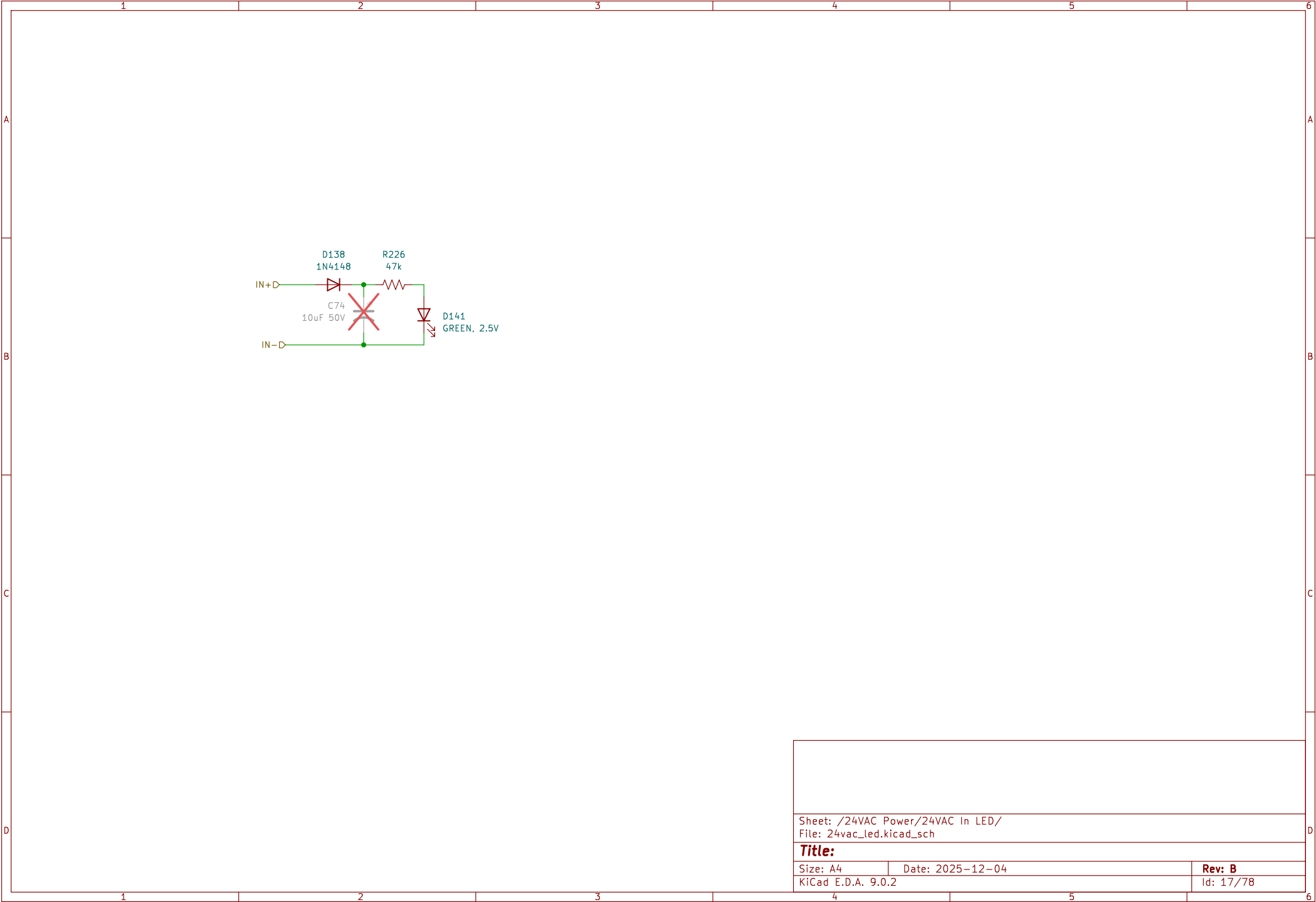
Size: A4

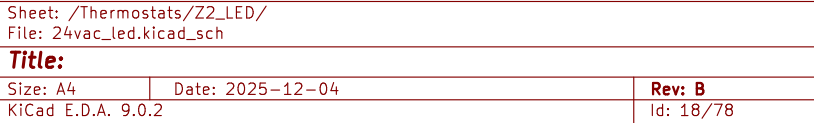
Date: 2025-12-04

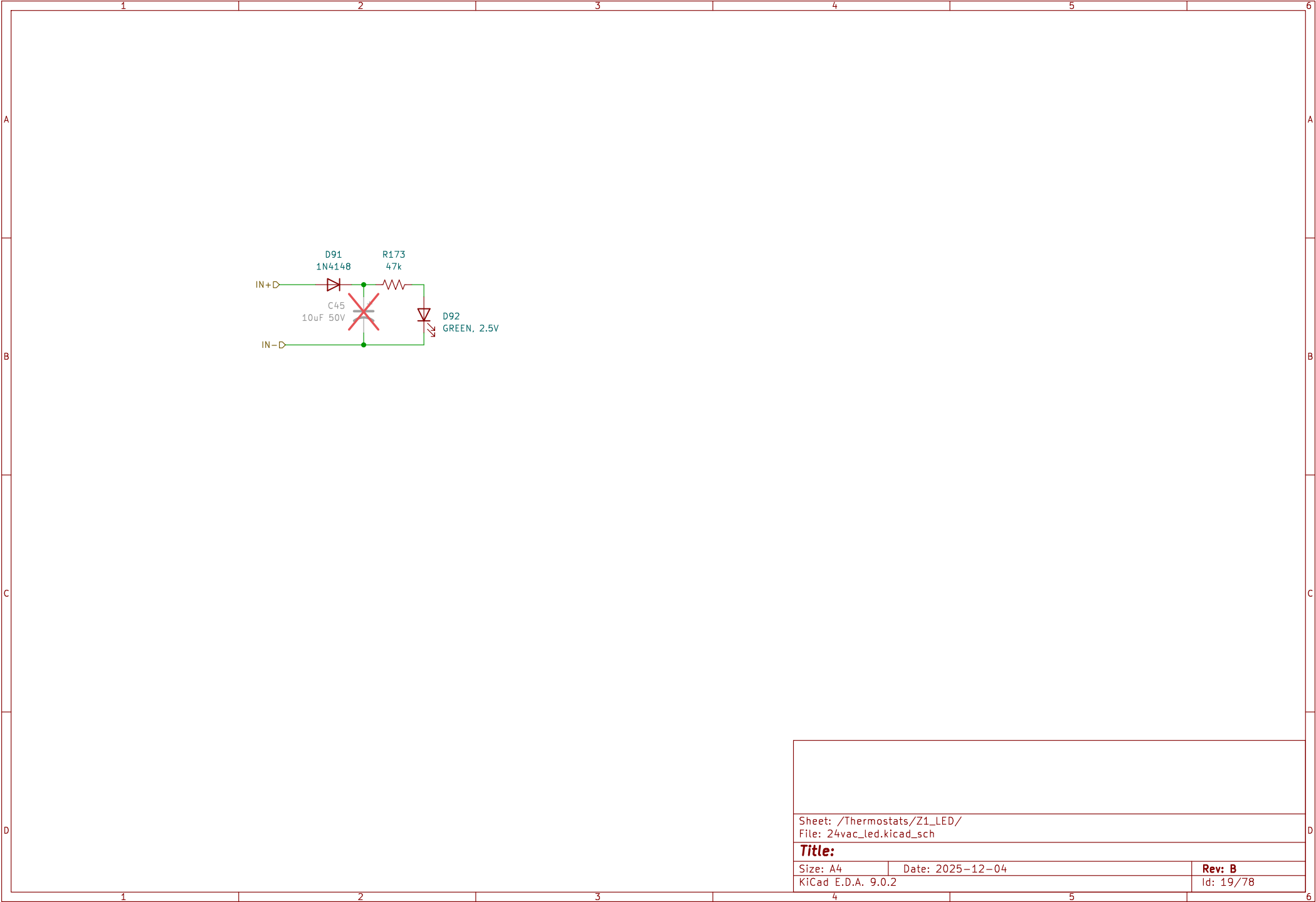
Rev: B

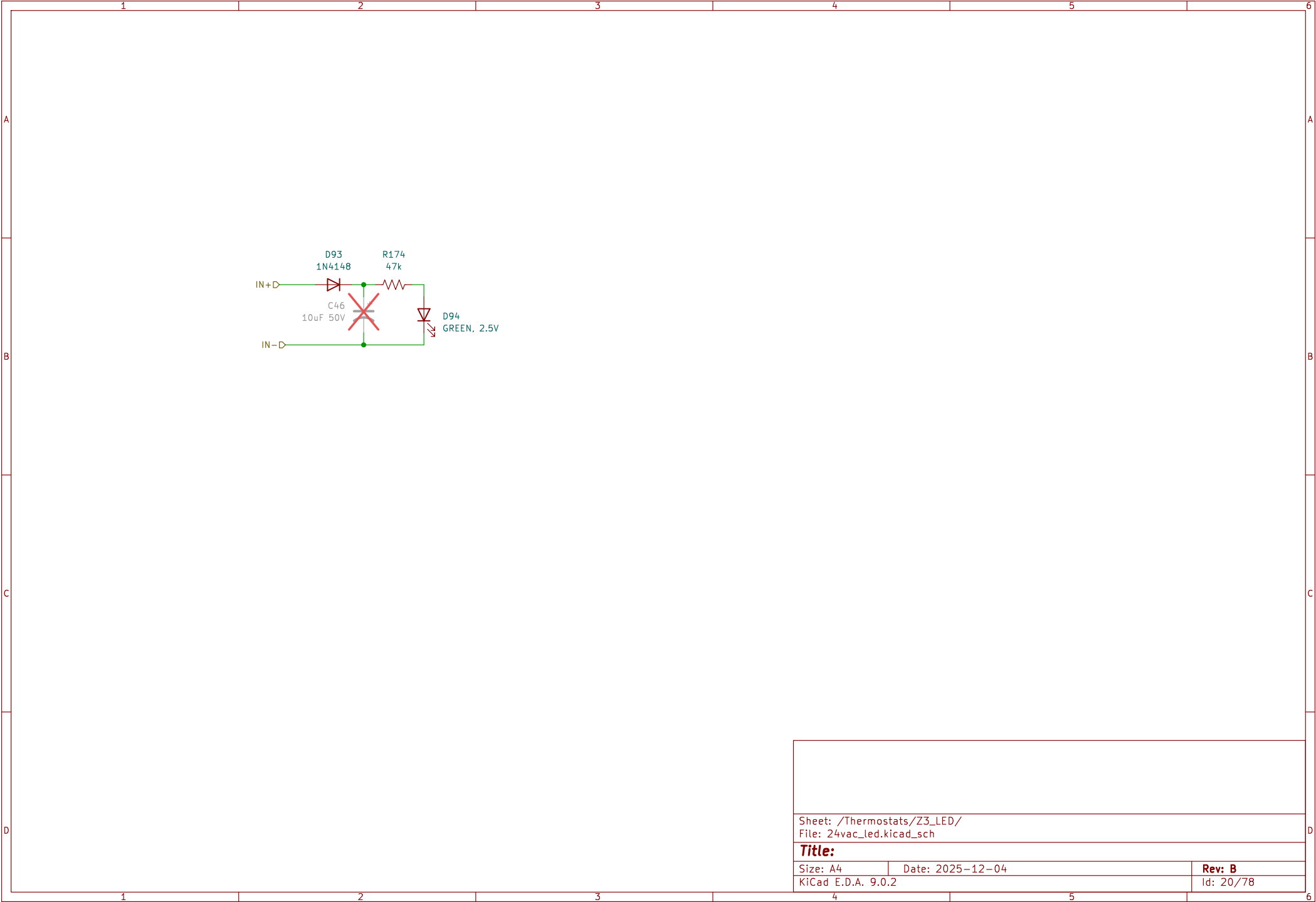
KiCad E.D.A. 9.0.2

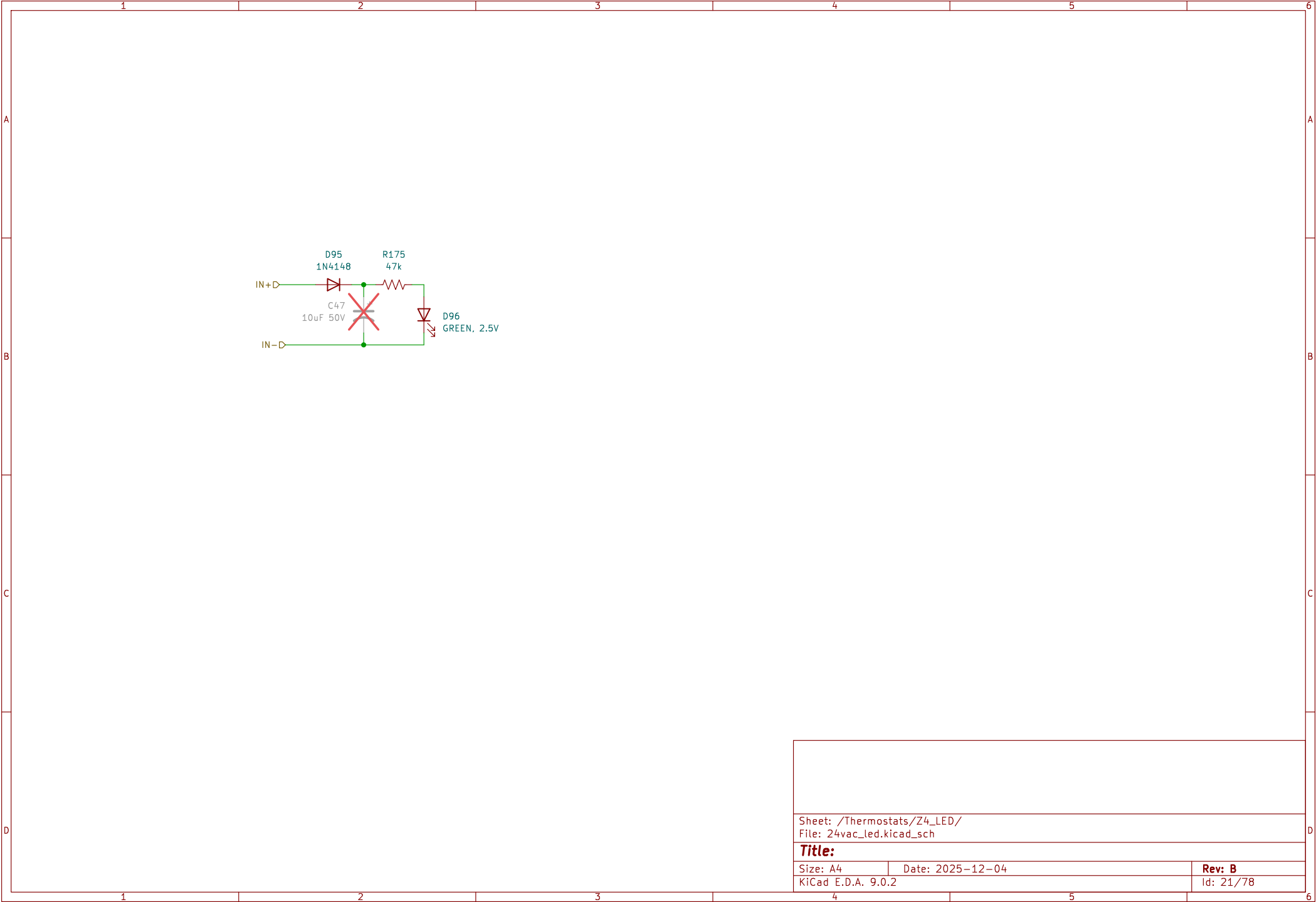
Id: 16/78

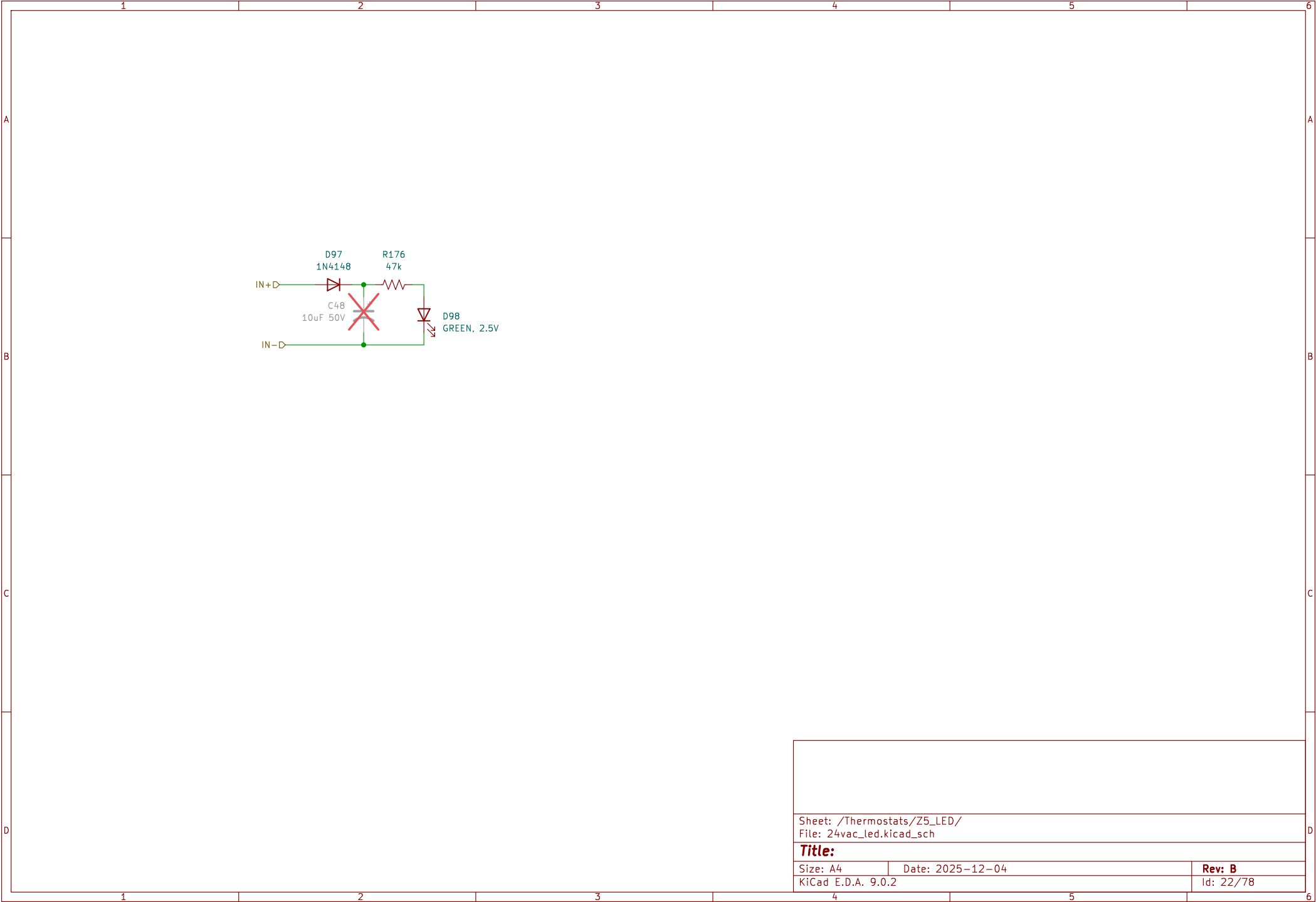


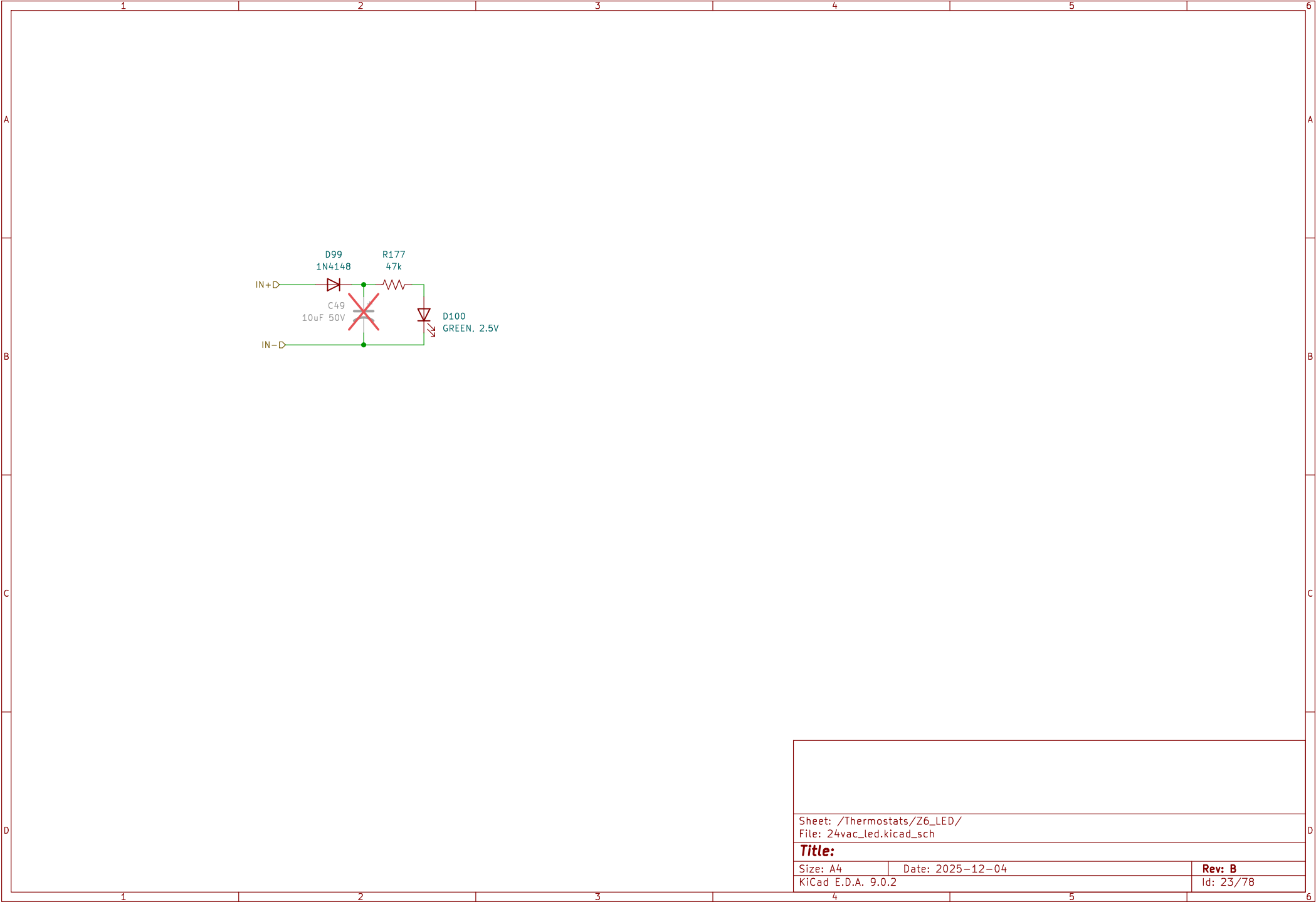


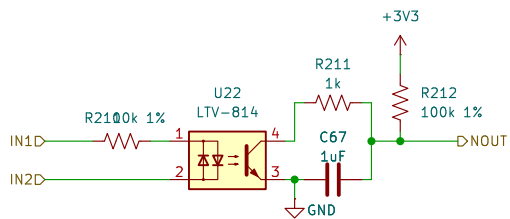












Sheet: /Thermostats/Z1_Optocoupler/
File: 24vac_optocoupler.kicad_sch

Title:

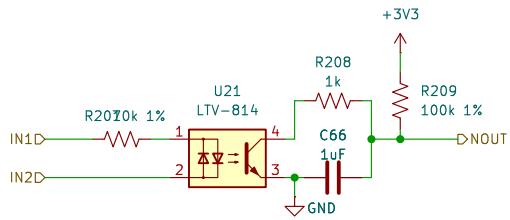
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 24/78



Sheet: /Thermostats/Z2_Optocoupler/
File: 24vac_optocoupler.kicad_sch

Title:

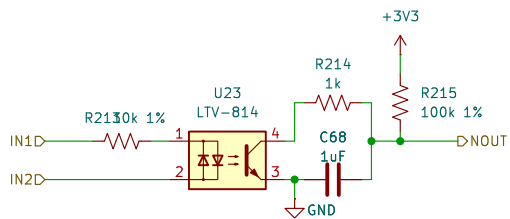
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 25/78



Sheet: /Thermostats/Z3_Optocoupler/
File: 24vac_optocoupler.kicad_sch

Title:

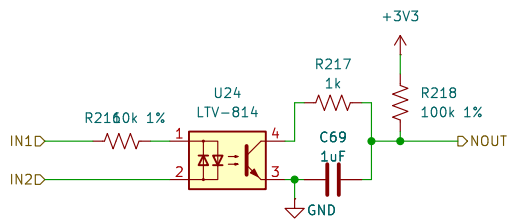
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 26/78



Sheet: /Thermostats/Z4_Optocoupler/
File: 24vac_optocoupler.kicad_sch

Title:

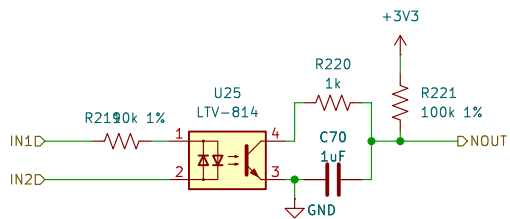
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 27/78



Sheet: /Thermostats/Z5_Optocoupler/
File: 24vac_optocoupler.kicad_sch

Title:

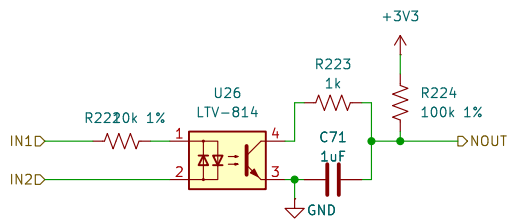
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 28/78



Sheet: /Thermostats/Z6_Optocoupler/
File: 24vac_optocoupler.kicad_sch

Title:

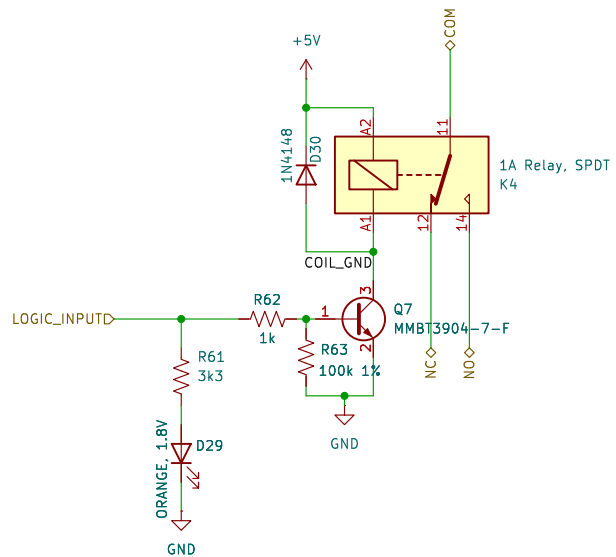
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 29/78



Sheet: /Zone Relays/Z1_FailSafe/
File: signal_relay.kicad_sch

Title:

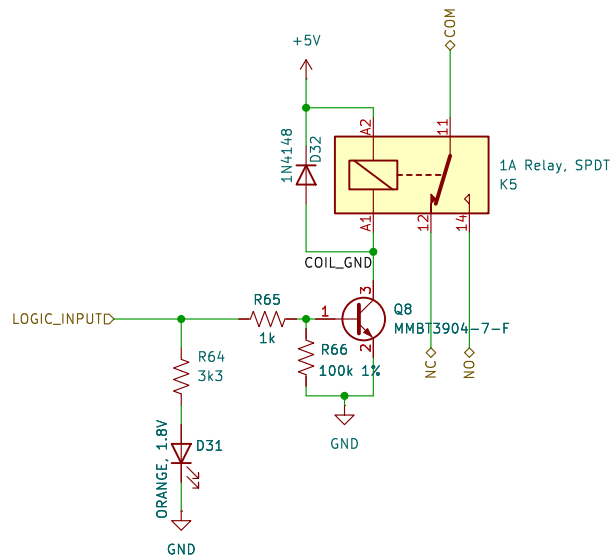
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 30/78



Sheet: /Zone Relays/Z1_OnOff/
File: signal_relay.kicad_sch

Title:

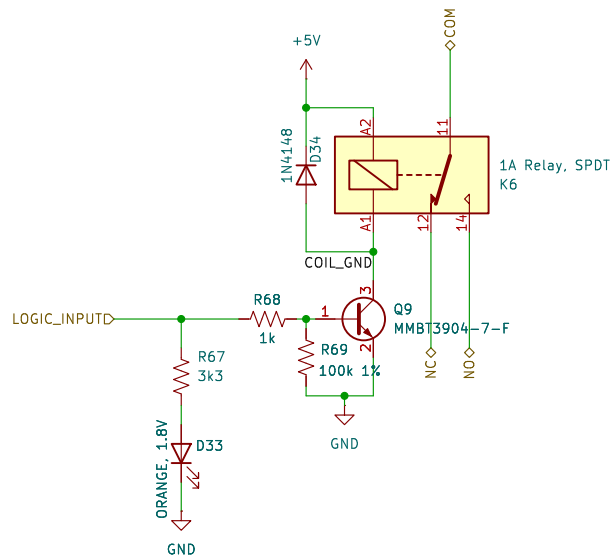
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 31/78



Sheet: /Zone Relays/Z2_FailSafe/
File: signal_relay.kicad_sch

Title:

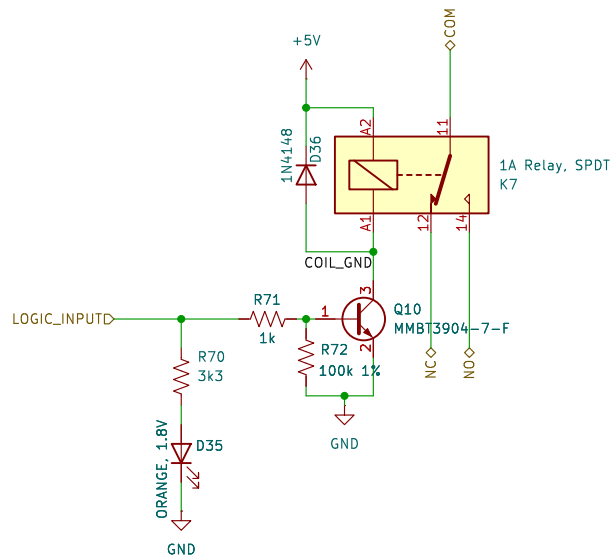
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 32/78



Sheet: /Zone Relays/Z2_OnOff/
File: signal_relay.kicad_sch

Title:

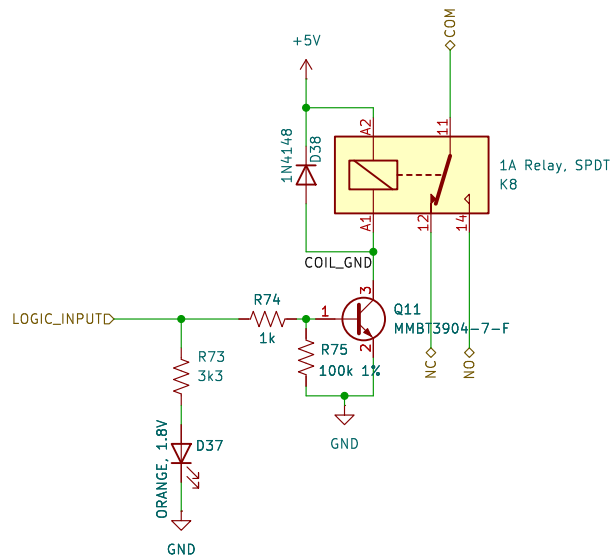
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 33/78



Sheet: /Zone Relays/Z3_FailSafe/
File: signal_relay.kicad_sch

Title:

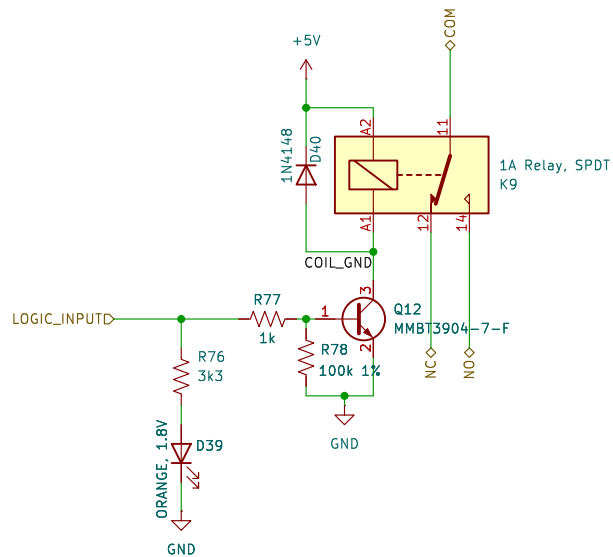
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 34/78



Sheet: /Zone Relays/Z3_OnOff/
File: signal_relay.kicad_sch

Title:

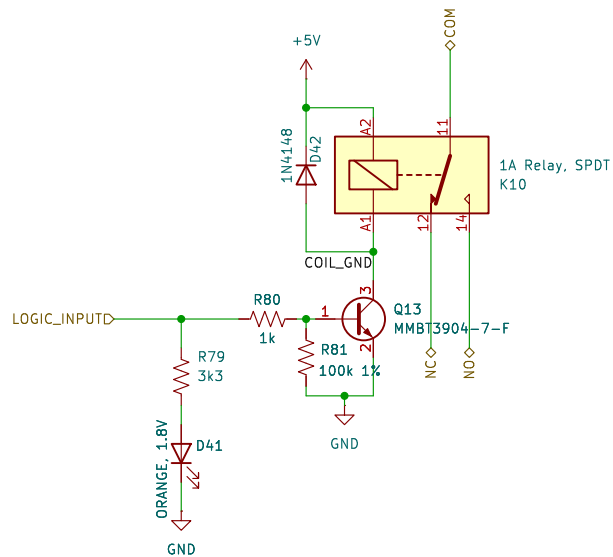
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 35/78



Sheet: /Zone Relays/Z4_OnOff/
File: signal_relay.kicad_sch

Title:

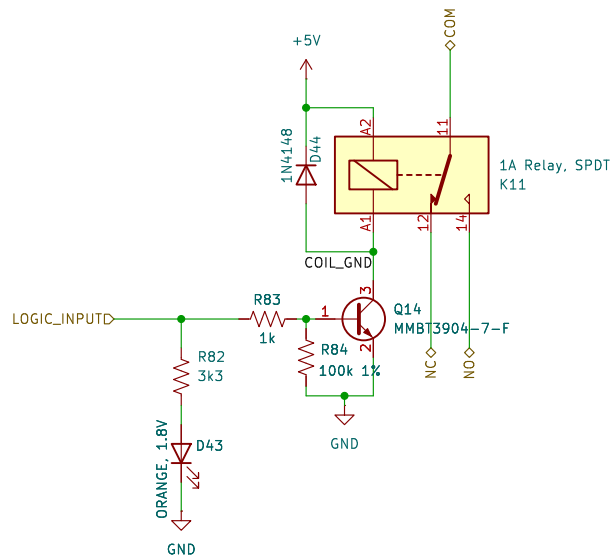
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 36/78



Sheet: /Zone Relays/Z4_FailSafe/
File: signal_relay.kicad_sch

Title:

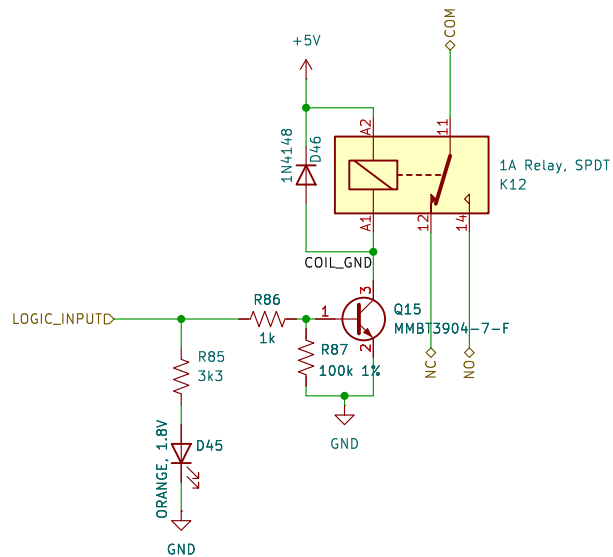
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 37/78



Sheet: /Zone Relays/Z5_FailSafe/
File: signal_relay.kicad_sch

Title:

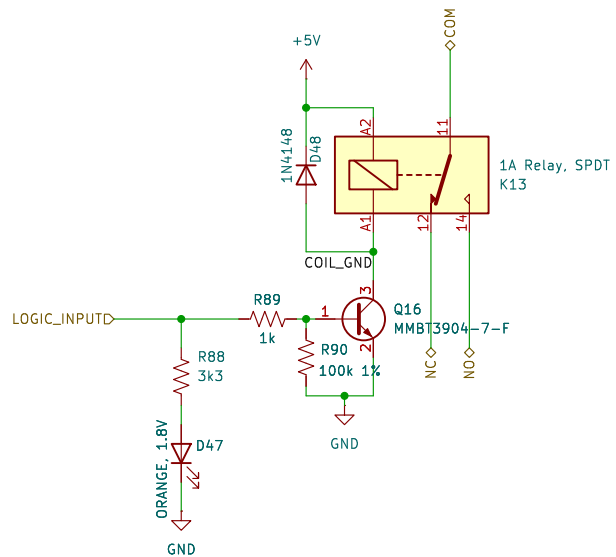
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 38/78



Sheet: /Zone Relays/Z5_OnOff/
File: signal_relay.kicad_sch

Title:

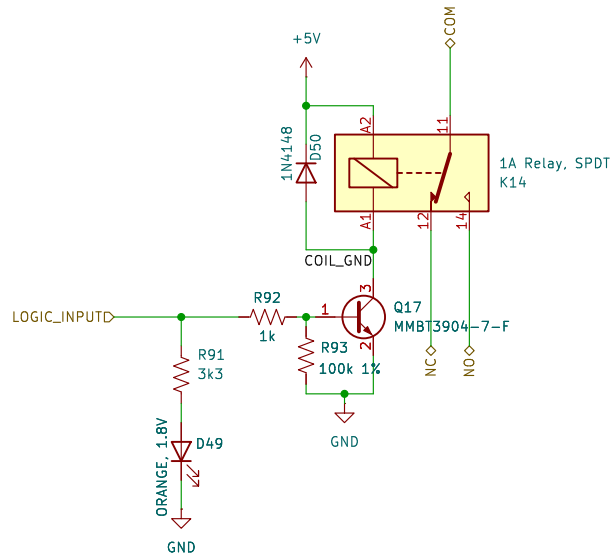
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 39/78



Sheet: /Zone Relays/Z6_FailSafe/
File: signal_relay.kicad_sch

Title:

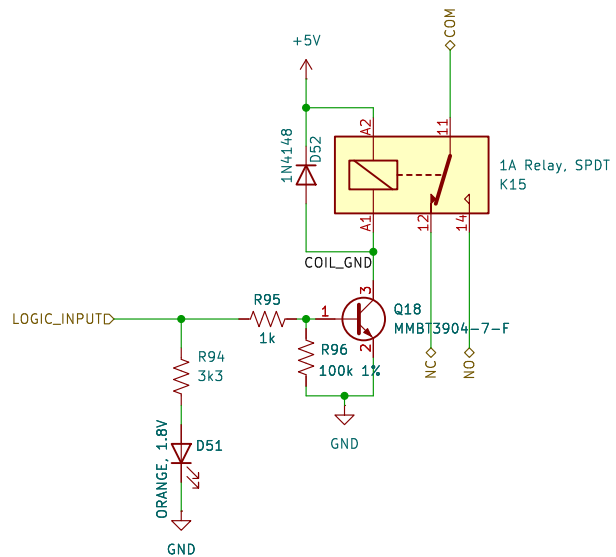
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 40/78



Sheet: /Zone Relays/Z6_OnOff/
File: signal_relay.kicad_sch

Title:

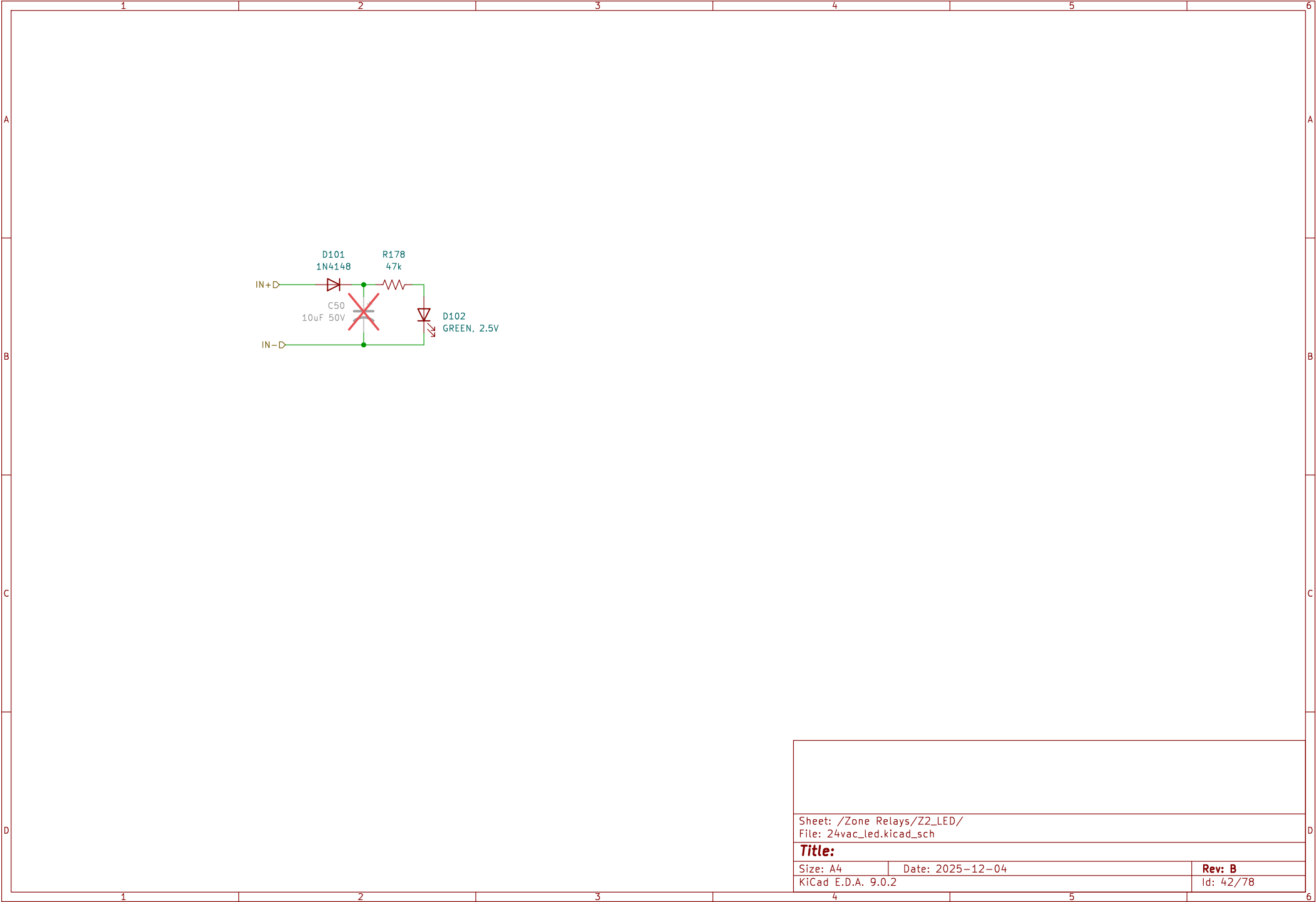
Size: A4

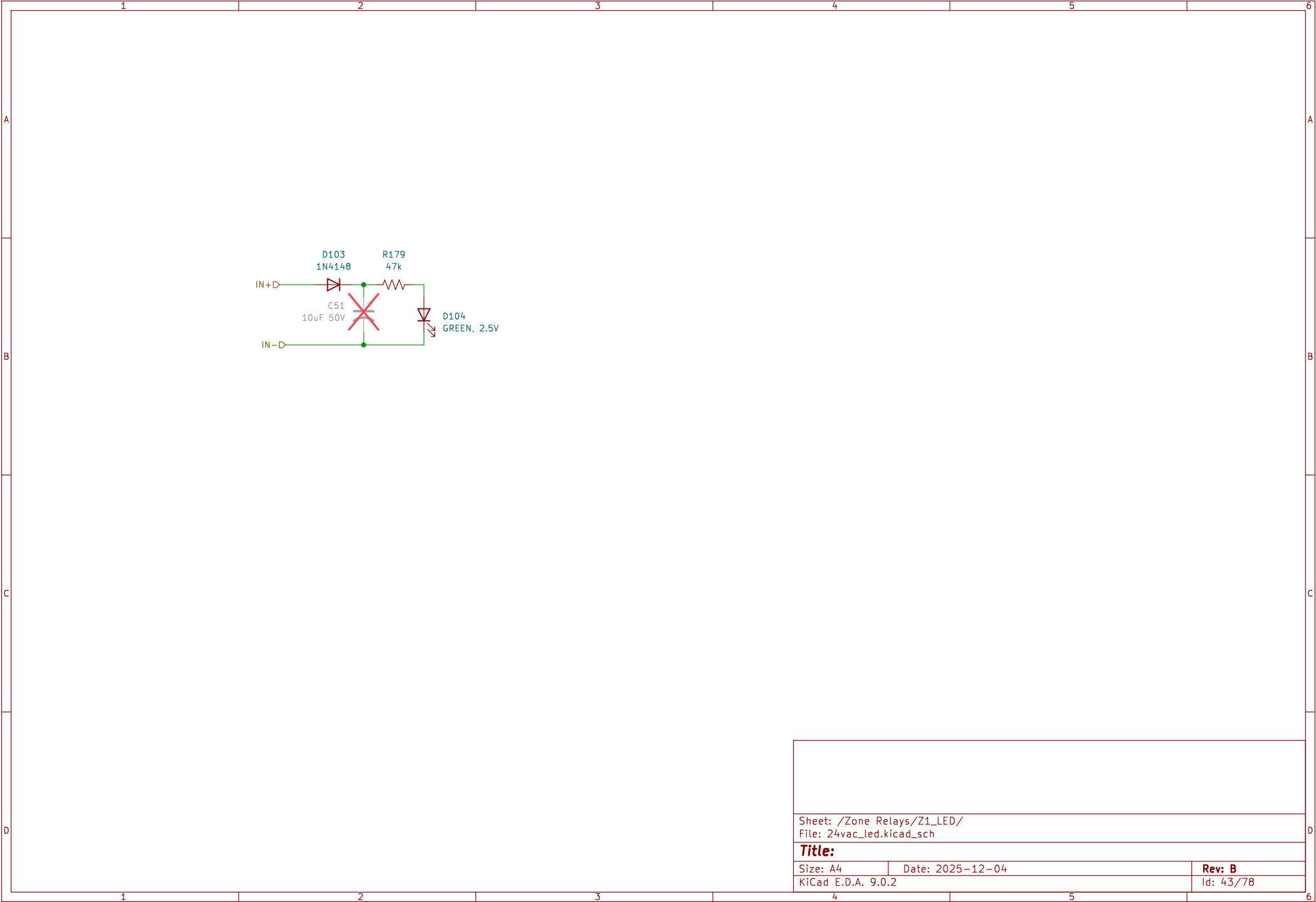
Date: 2025-12-04

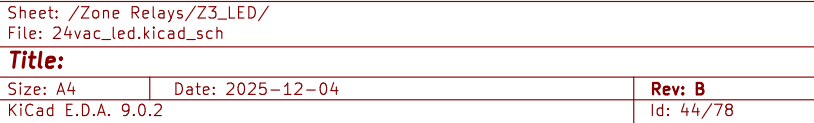
Rev: B

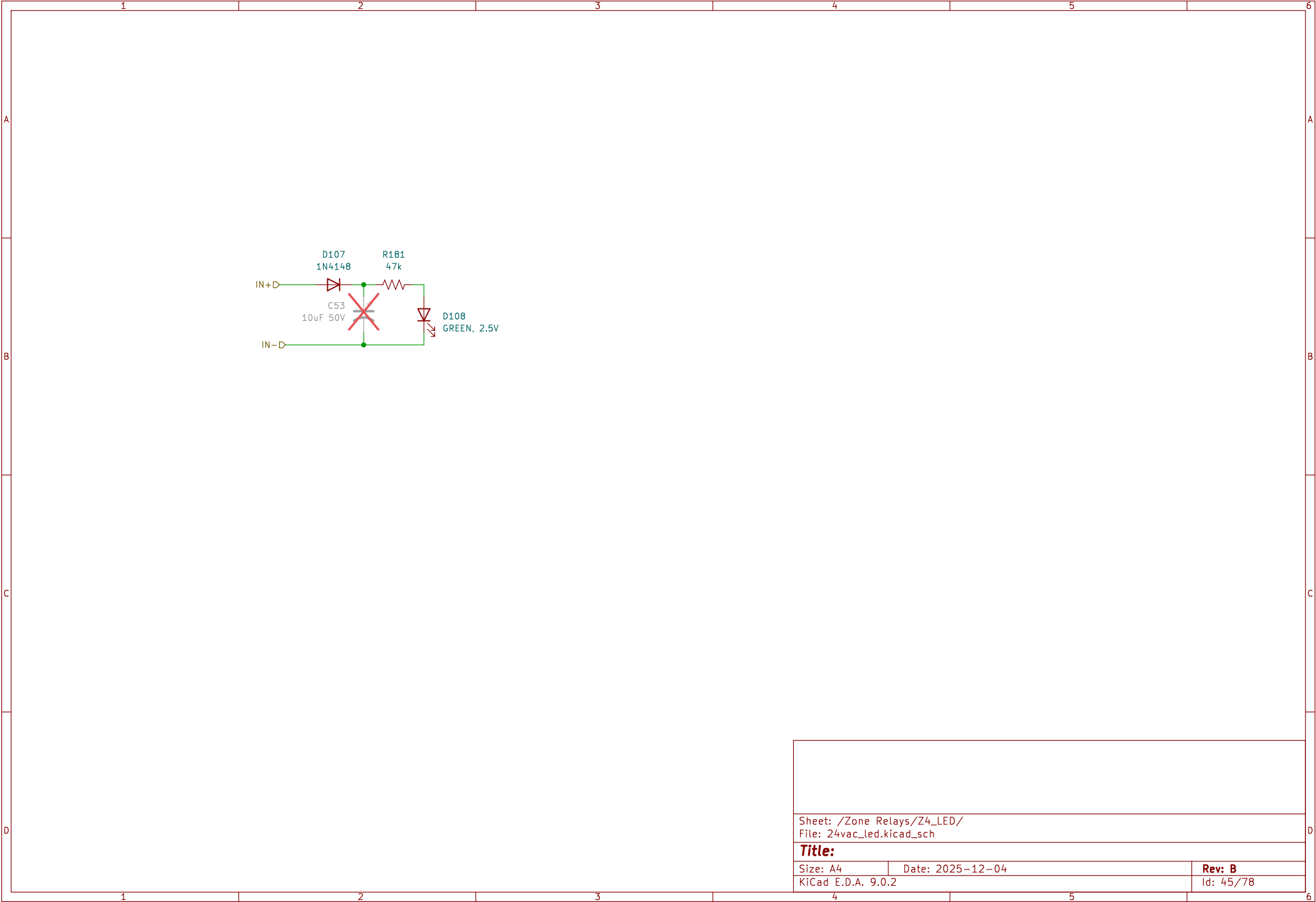
KiCad E.D.A. 9.0.2

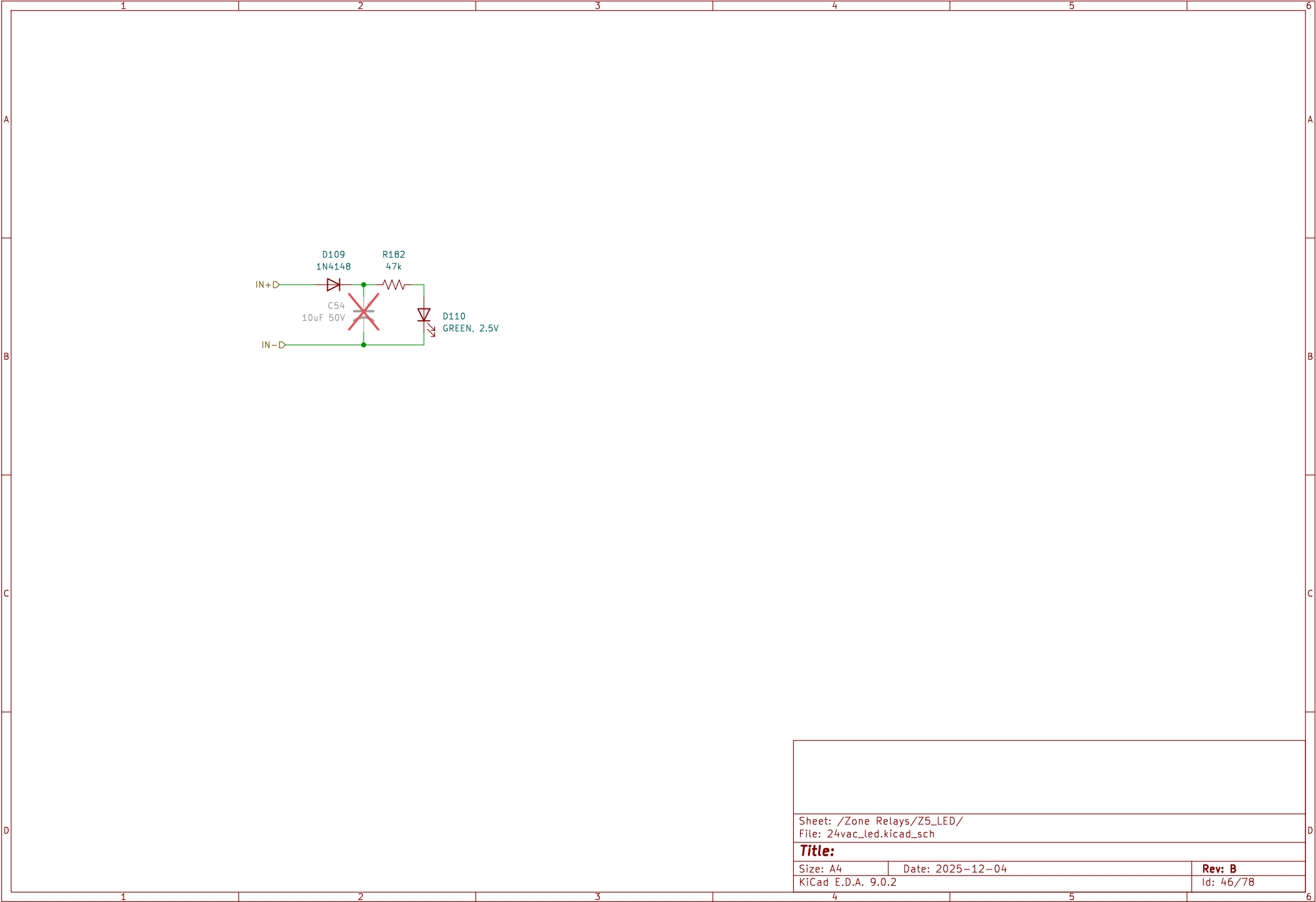
Id: 41/78



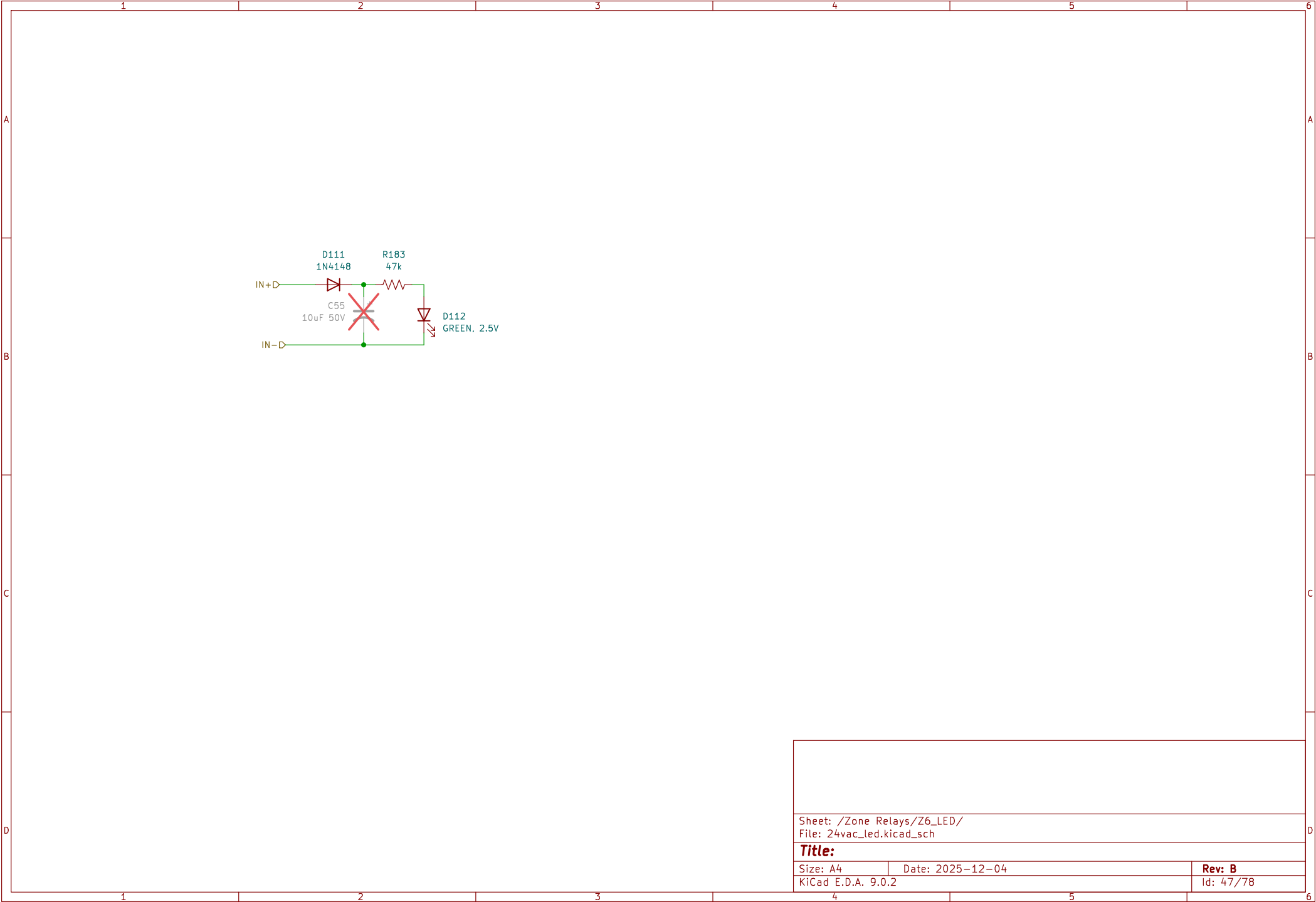


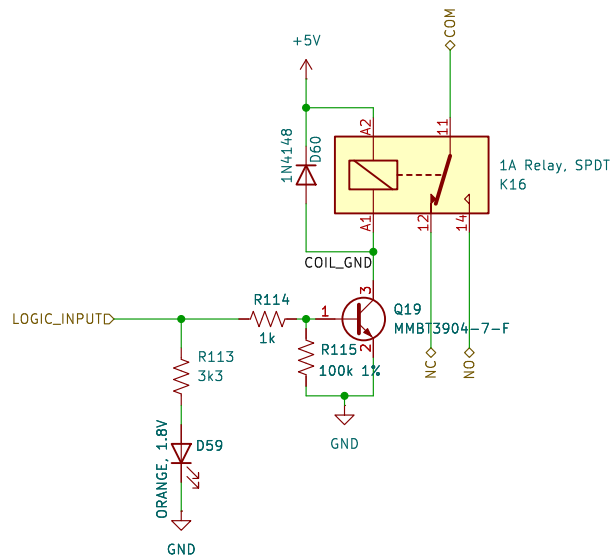






Sheet: /Zone Relays/Z5_LED/ File: 24vac_led.kicad_sch		
Title:		
Size: A4	Date: 2025-12-04	Rev: B
KiCad E.D.A. 9.0.2		Id: 46/78





Sheet: /System Outputs/Primary Pump Relay/
File: signal_relay.kicad_sch

Title:

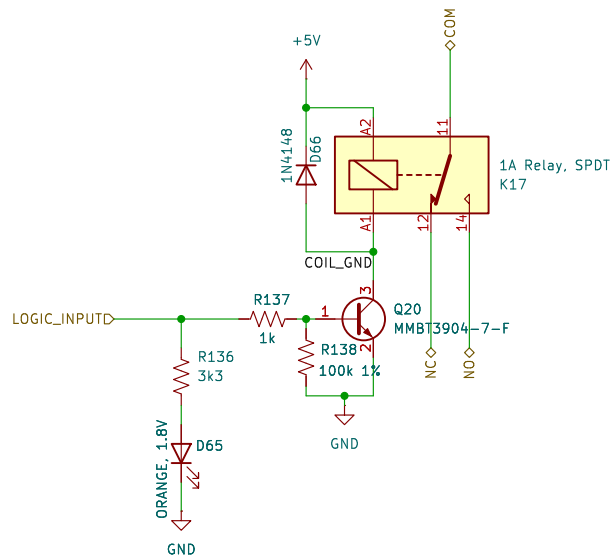
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 48/78



Sheet: /System Outputs/Heat Pump Dry Contact Relay/
File: signal_relay.kicad_sch

Title:

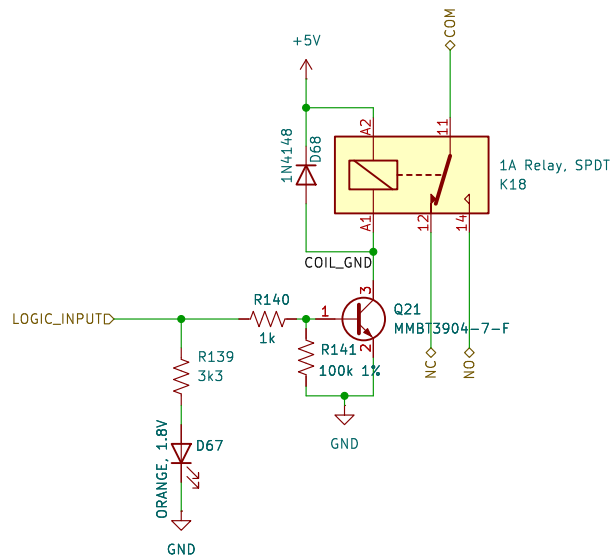
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 49/78



Sheet: /System Outputs/Buffer Tank Upper Contactor Relay/
File: signal_relay.kicad_sch

Title:

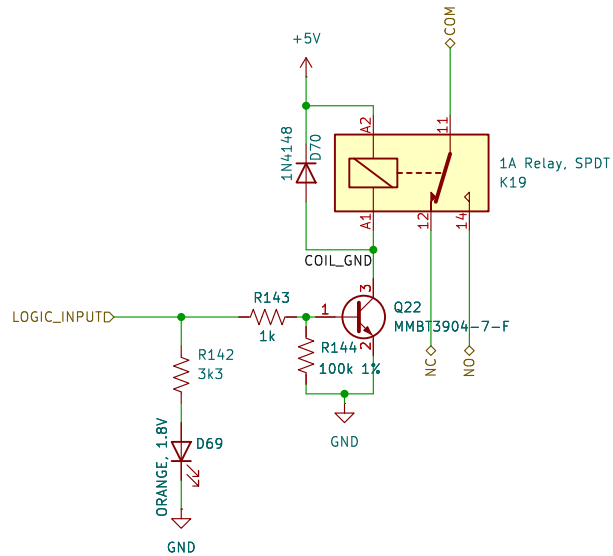
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 50/78



Sheet: /System Outputs/Buffer Tank Lower Contactor Relay/
File: signal_relay.kicad_sch

Title:

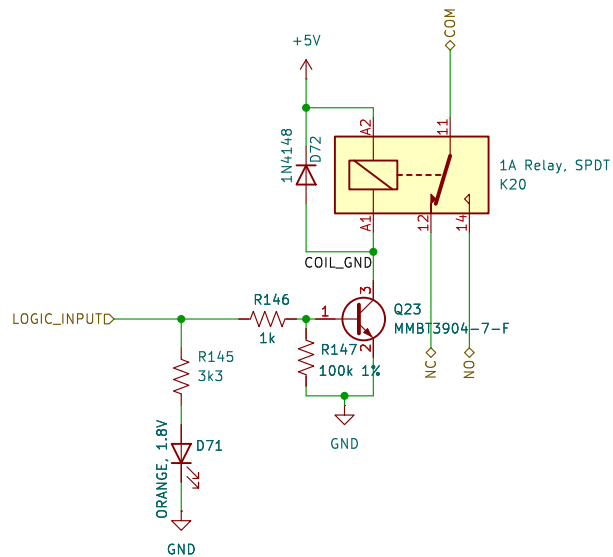
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 51/78



Sheet: /System Outputs/Boiler+Buffer Valves Relay/
File: signal_relay.kicad_sch

Title:

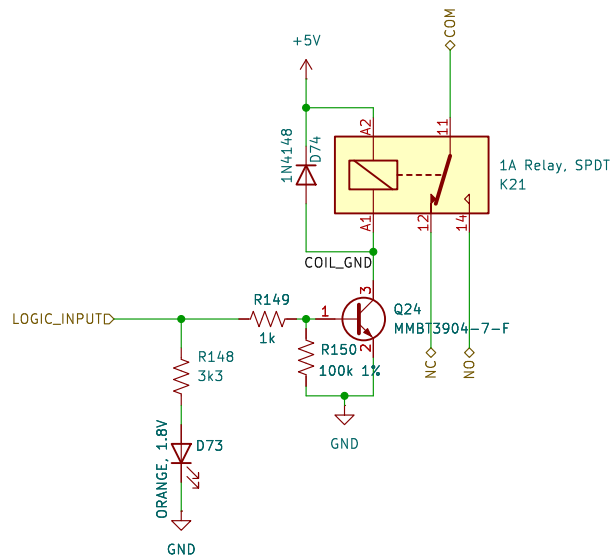
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 52/78



Sheet: /System Outputs/Boiler Intercept Relay/
File: signal_relay.kicad_sch

Title:

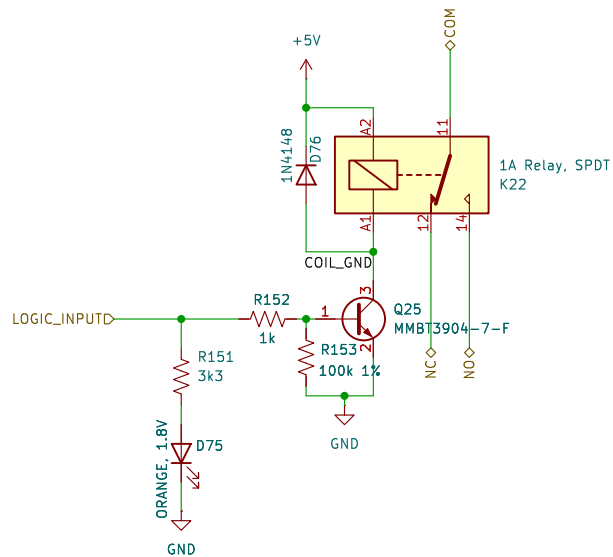
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 53/78



Sheet: /System Outputs/Misc Relay 1/
File: signal_relay.kicad_sch

Title:

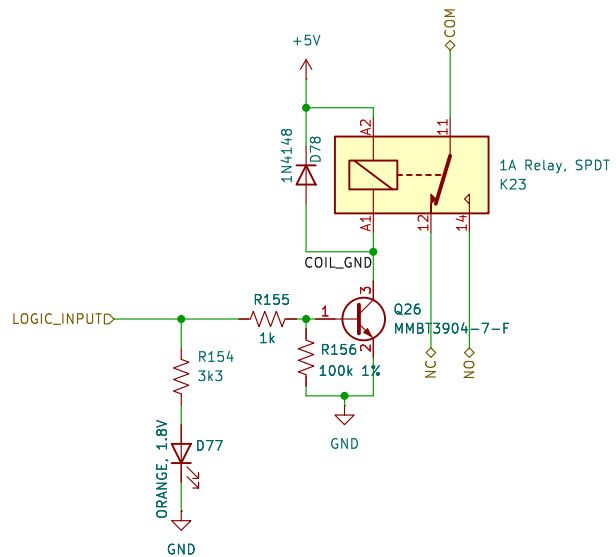
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 54/78



Sheet: /System Outputs/Misc Relay 2/
File: signal_relay.kicad_sch

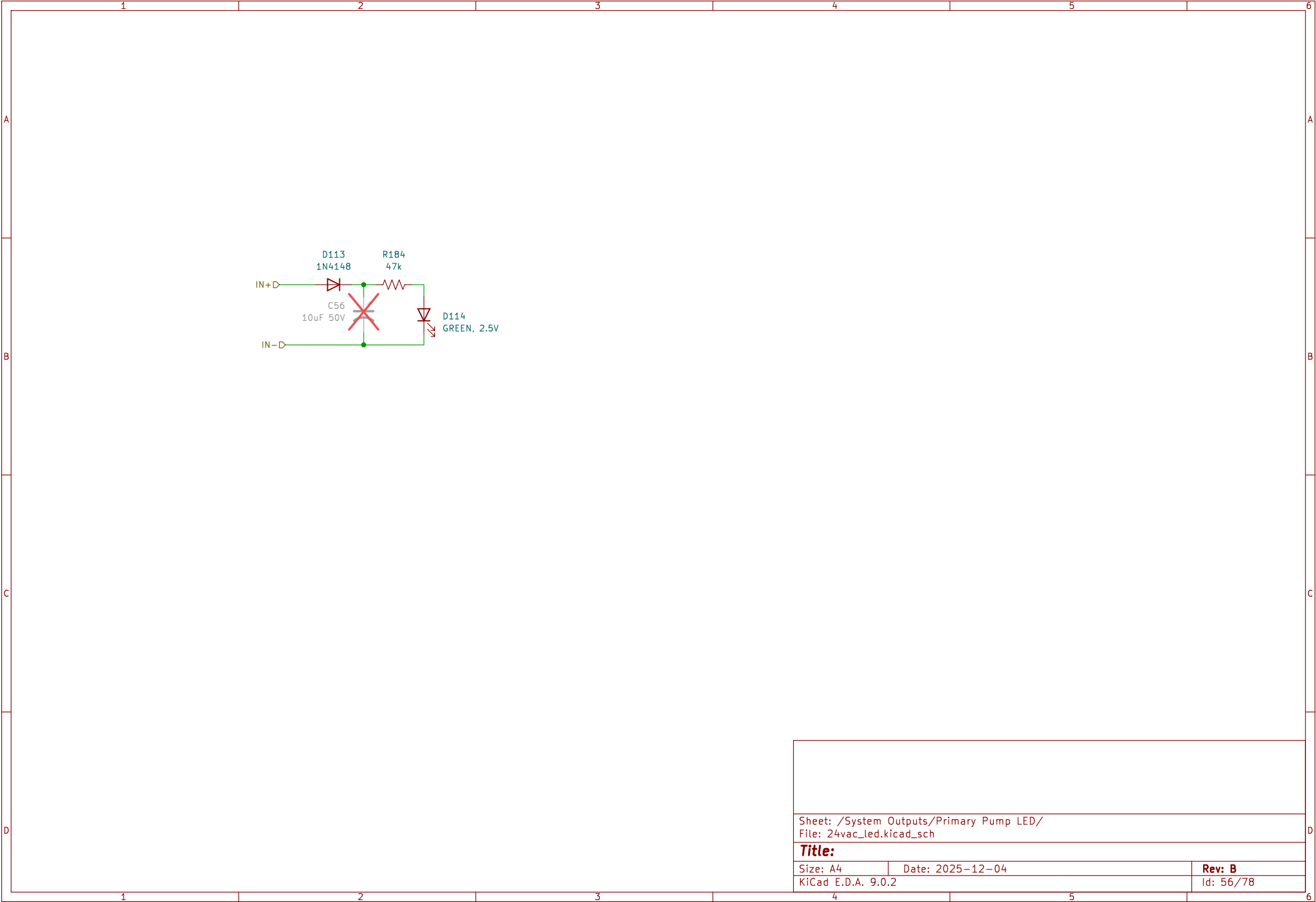
Title:

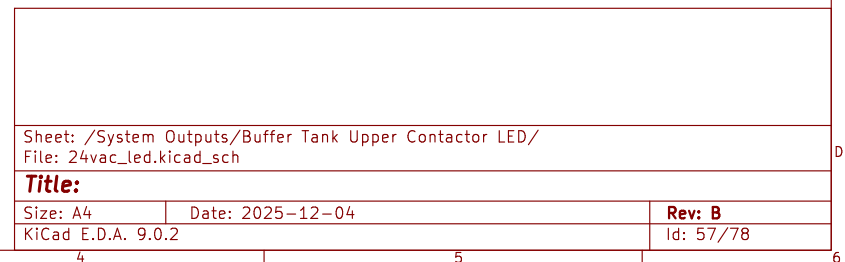
Size: A4
KiCad E.D.A. 9.0.2

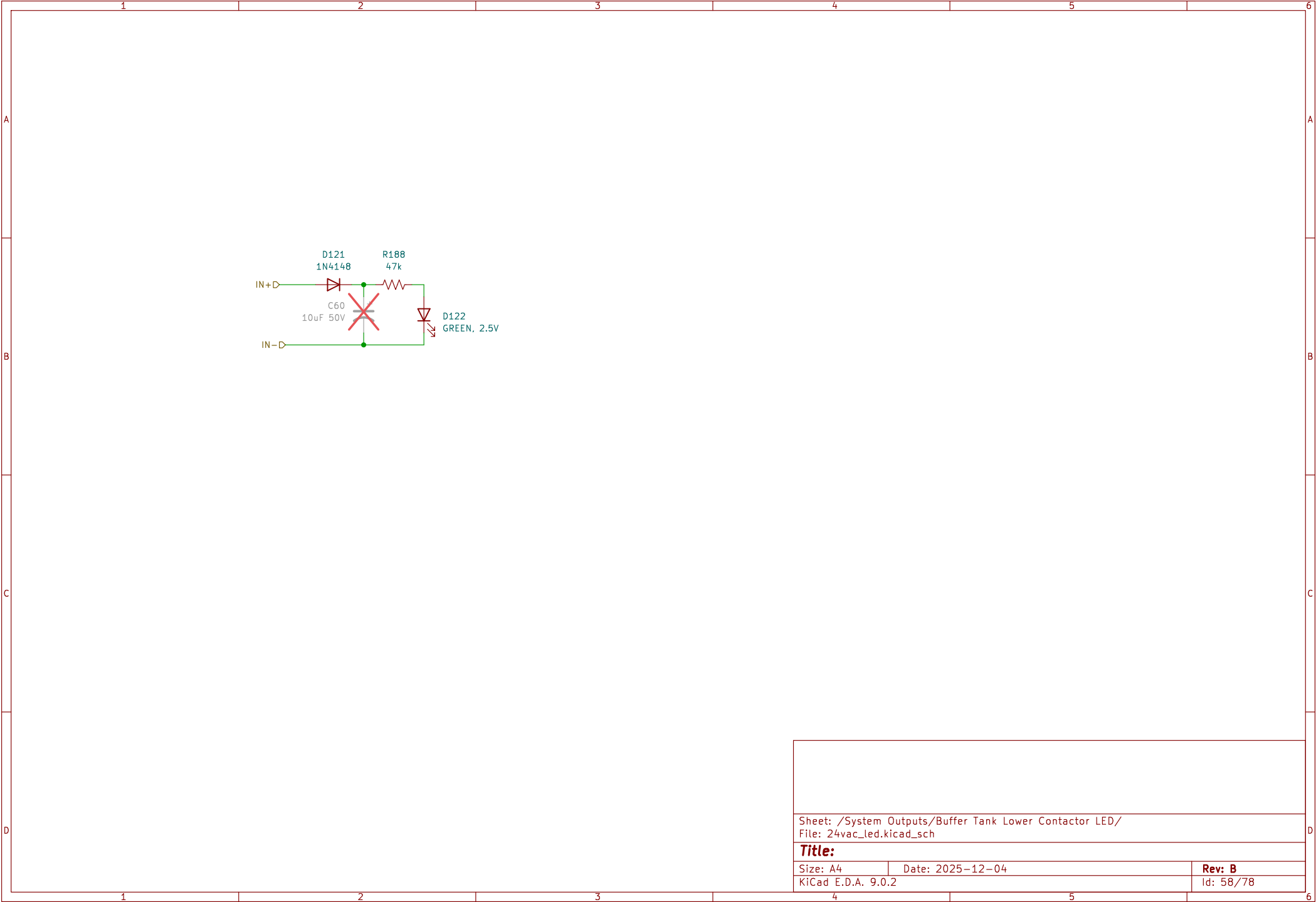
Date: 2025-12-04

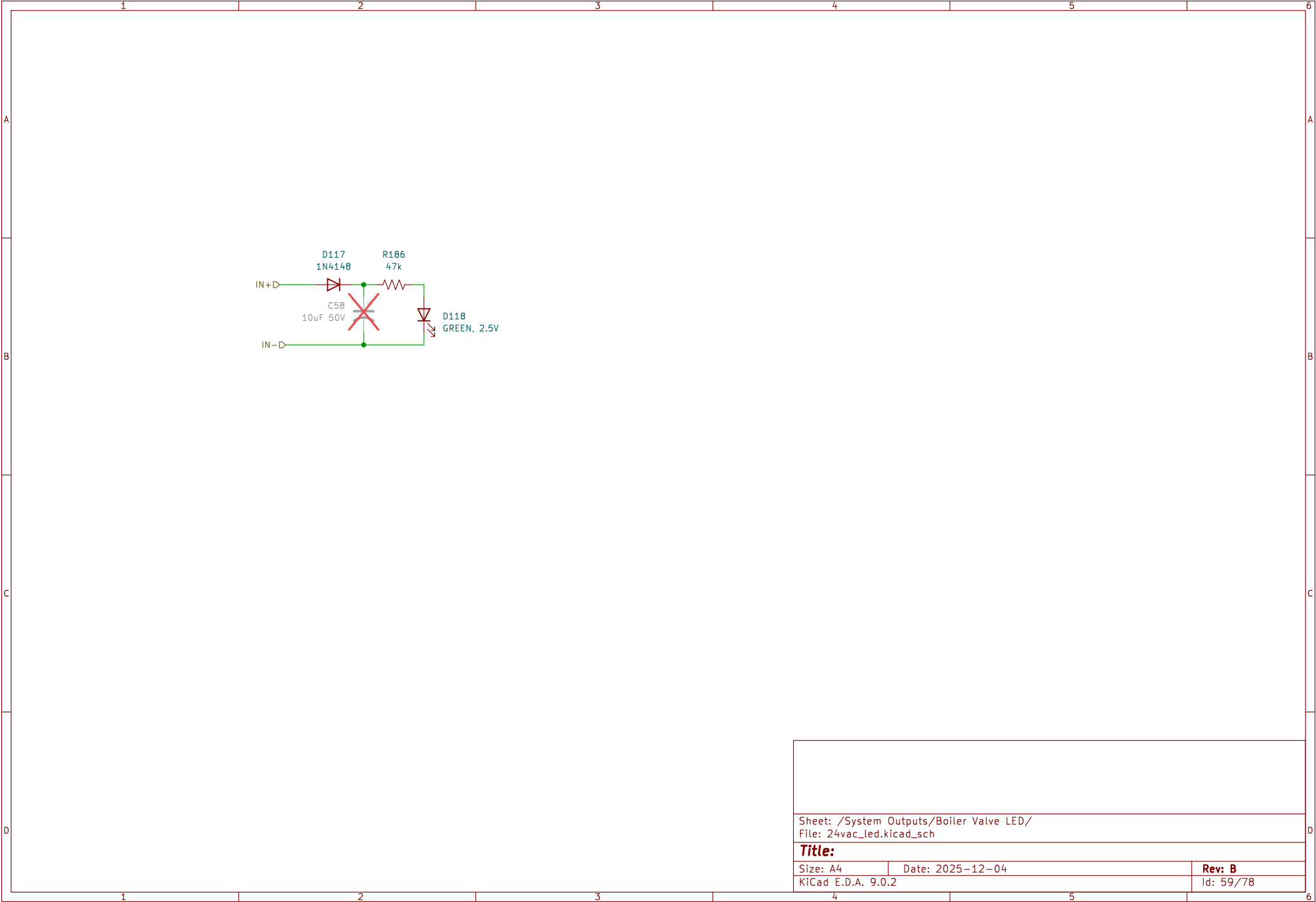
Rev: B

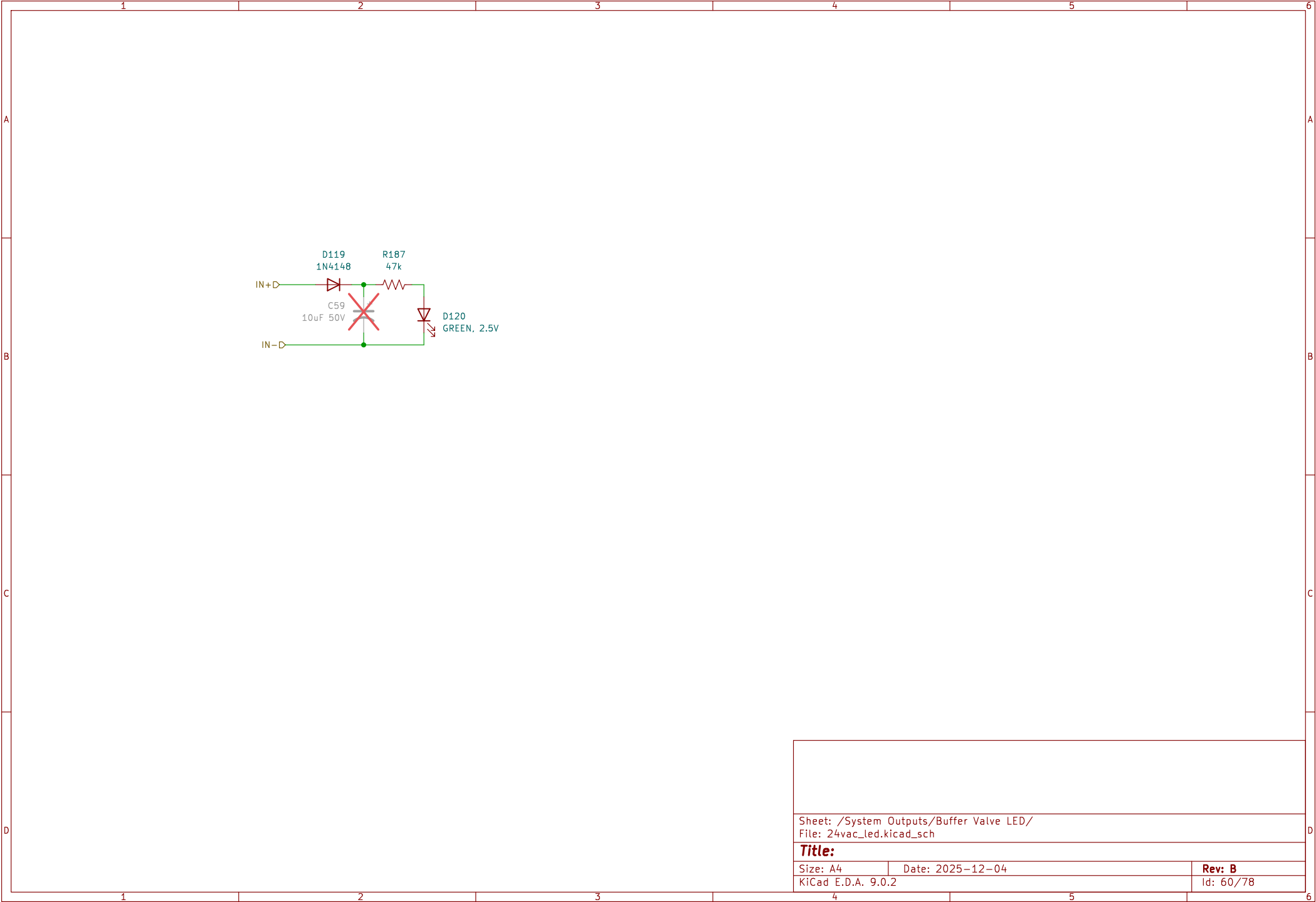
Id: 55/78

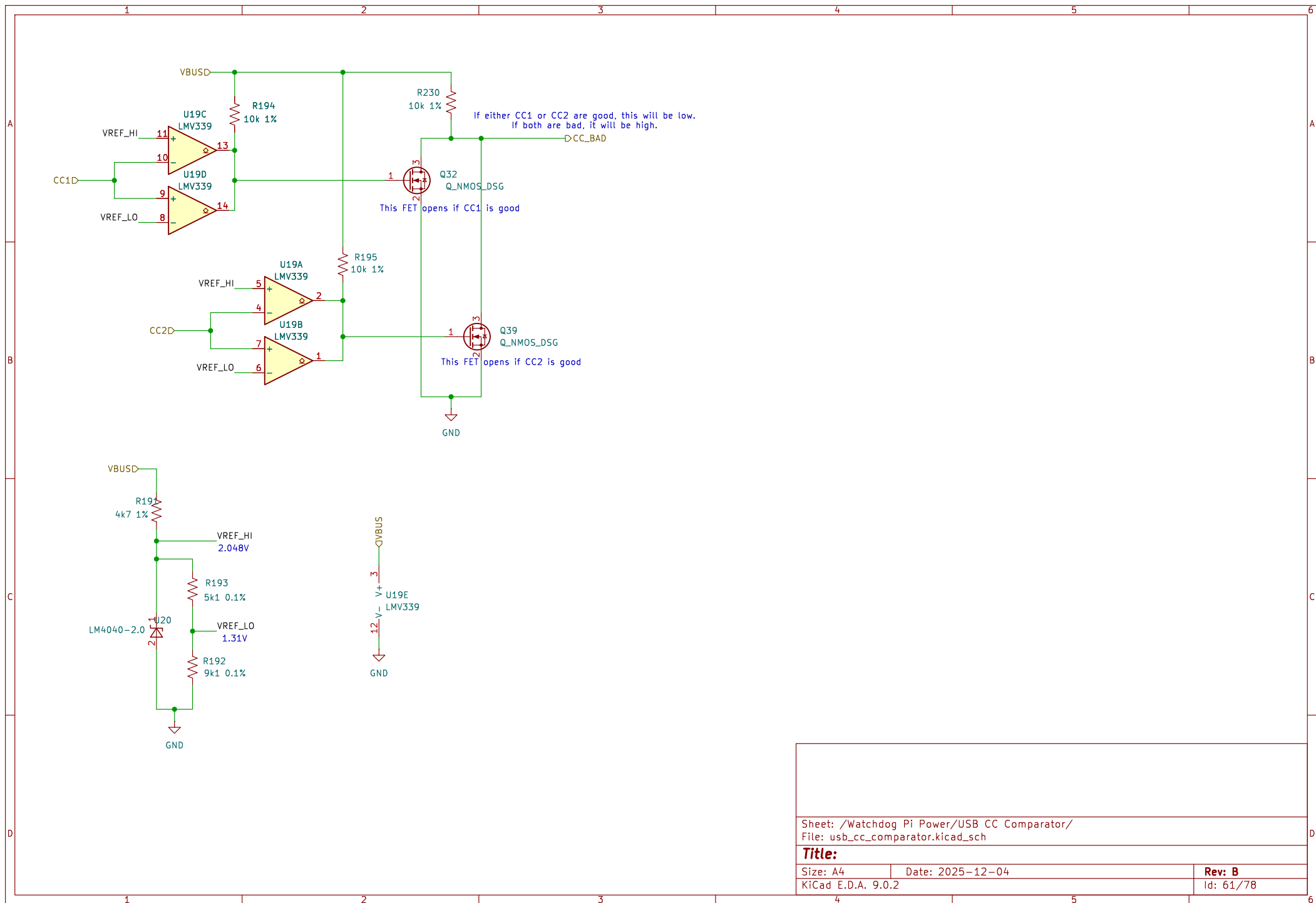












Sheet: /Watchdog PI Power/USB CC Comparator/
File: usb_cc_comparator.kicad_sch

Title:

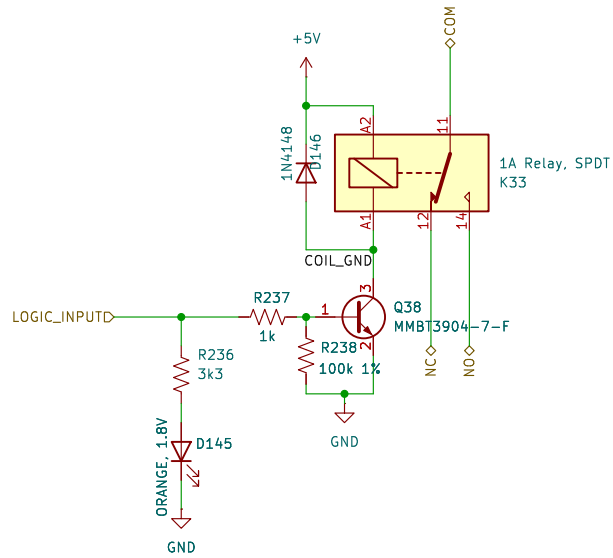
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 61/78



Sheet: /System Outputs/Store Tank Lower Contactor Relay/
File: signal_relay.kicad_sch

Title:

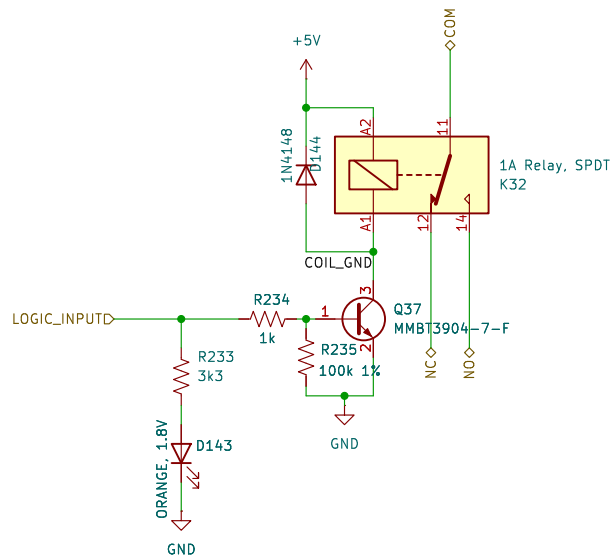
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 72/78



Sheet: /System Outputs/Store Tank Upper Contactor Relay/
File: signal_relay.kicad_sch

Title:

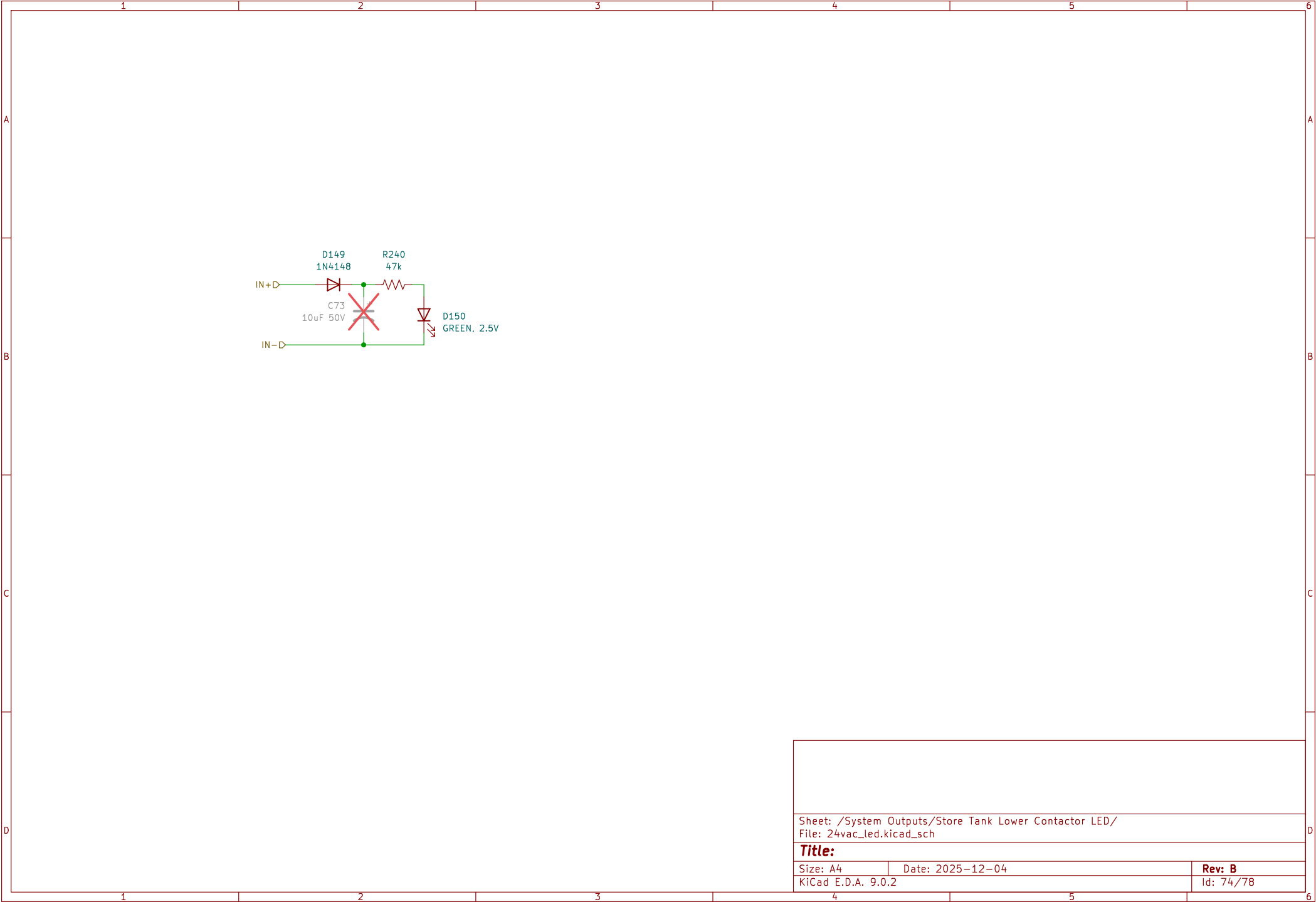
Size: A4

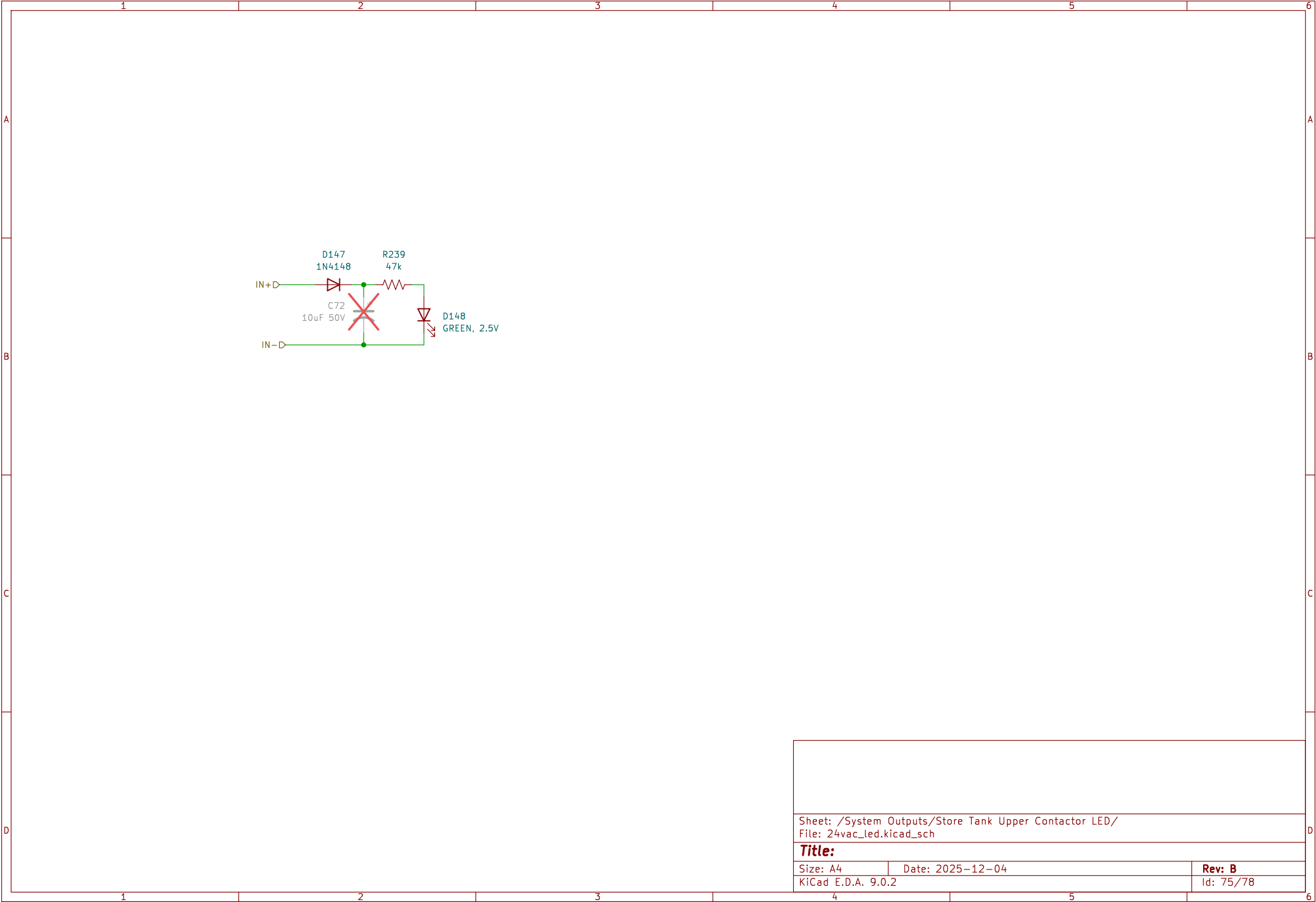
Date: 2025-12-04

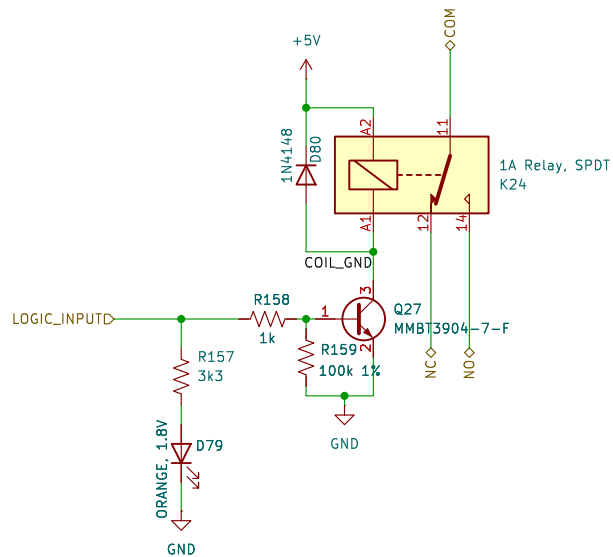
Rev: B

KiCad E.D.A. 9.0.2

Id: 73/78







Sheet: /FCM Outputs/Secondary Pump Relay/
File: signal_relay.kicad_sch

Title:

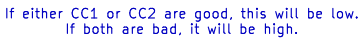
Size: A4

Date: 2025-12-04

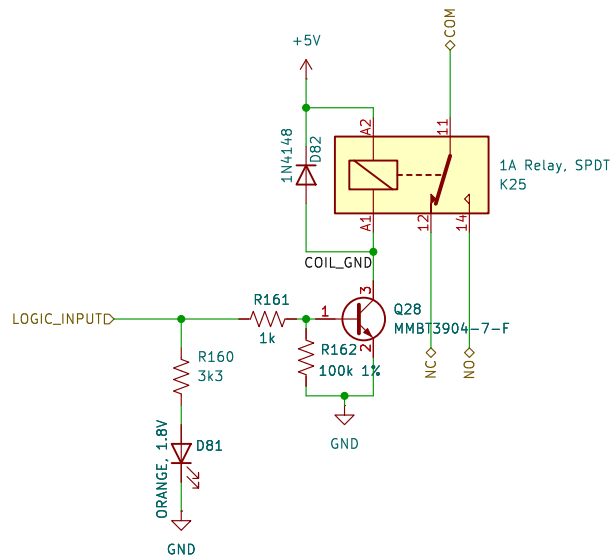
Rev: B

KiCad E.D.A. 9.0.2

Id: 76/78



Id: 77/78



Sheet: /FCM Outputs/Store Pump Relay/
File: signal_relay.kicad_sch

Title:

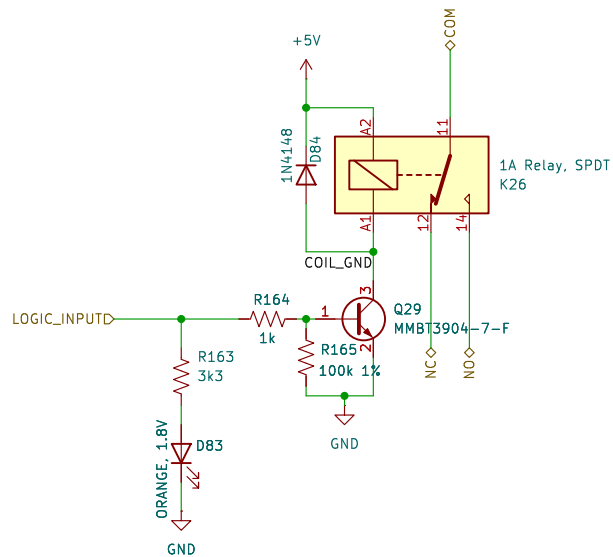
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 77/78



Sheet: /FCM Outputs/Discharge Valve Relay/
File: signal_relay.kicad_sch

Title:

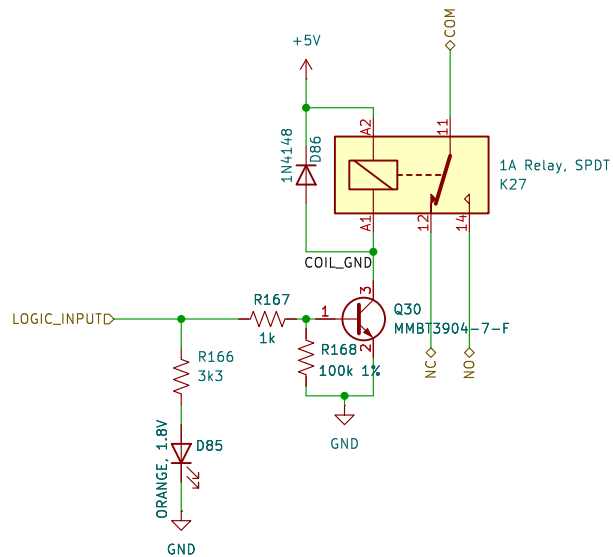
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 78/78



Sheet: /FCM Outputs/Iso Valve Relay/
File: signal_relay.kicad_sch

Title:

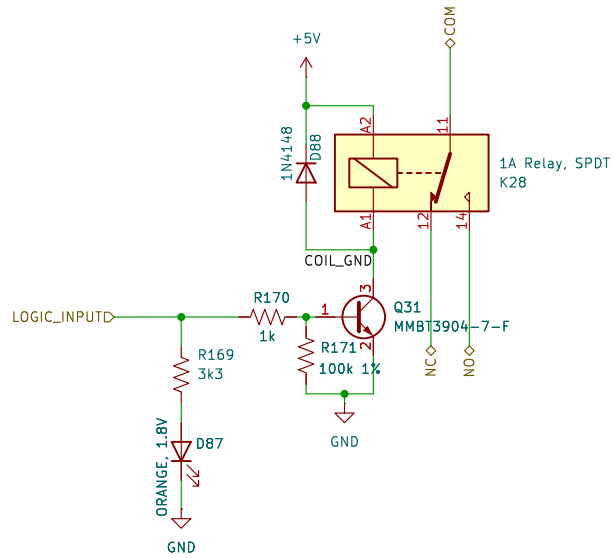
Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 79/78



Sheet: /FCM Outputs/Iso Failsafe Valve Relay/
File: signal_relay.kicad_sch

Title:

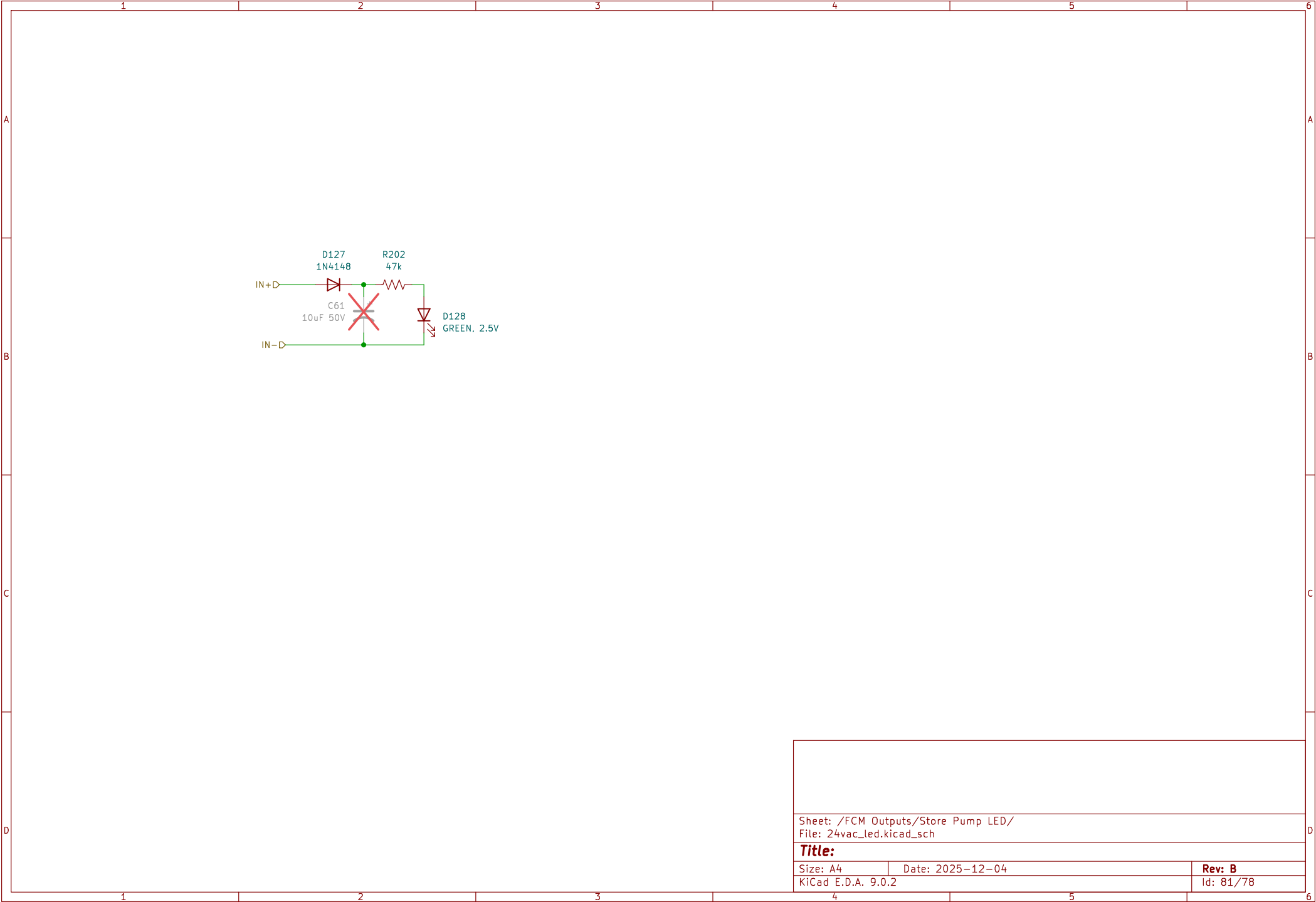
Size: A4

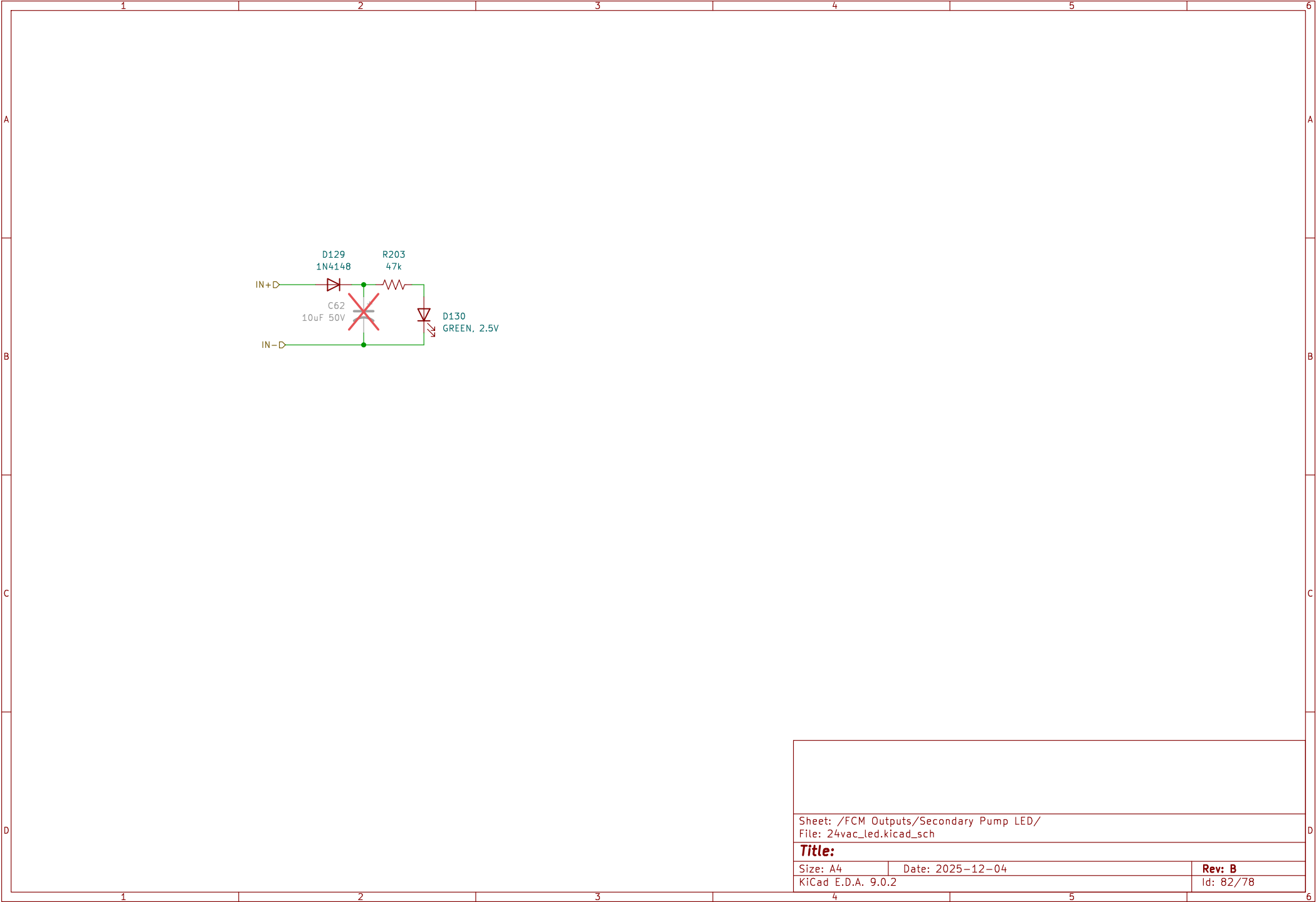
Date: 2025-12-04

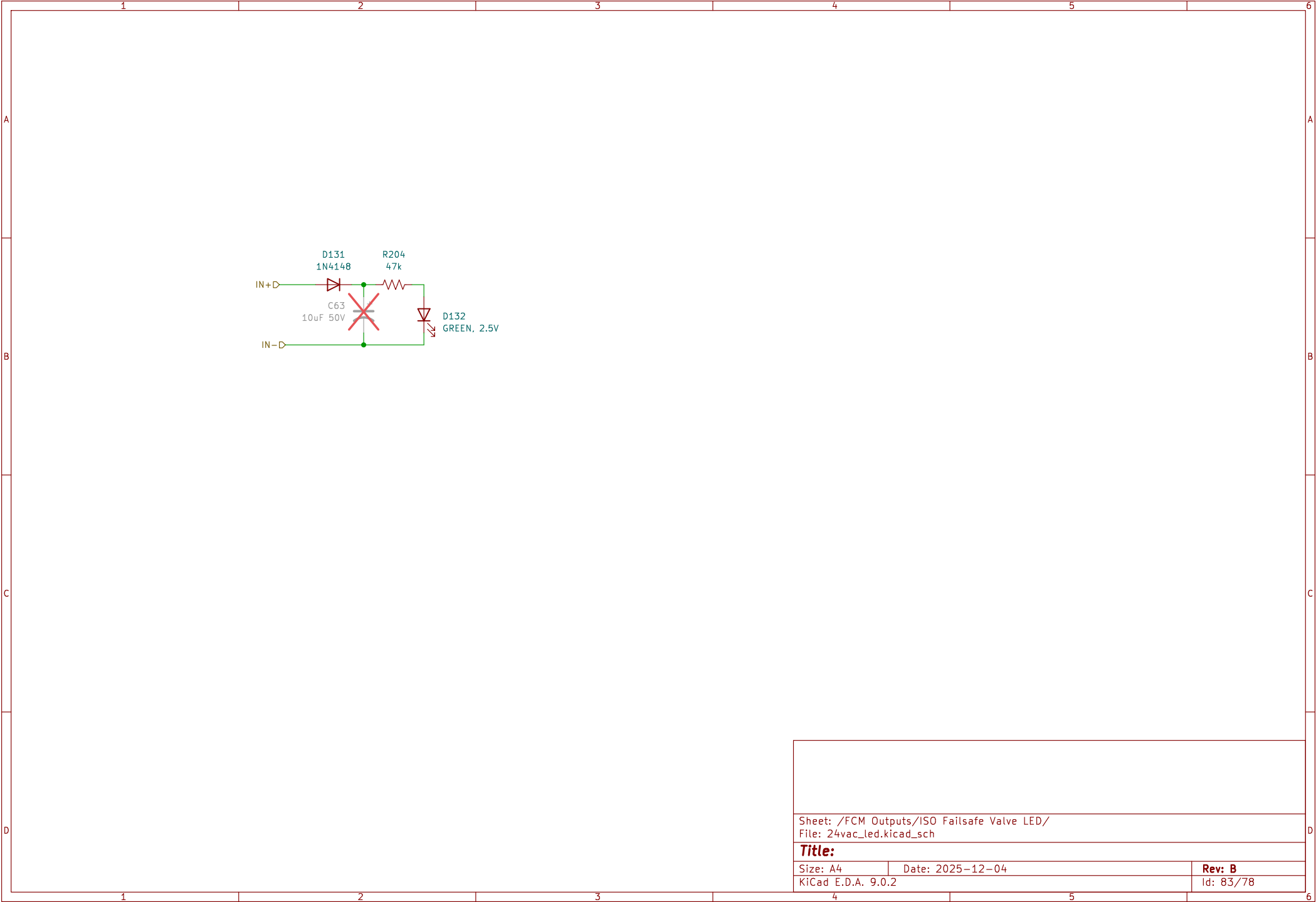
Rev: B

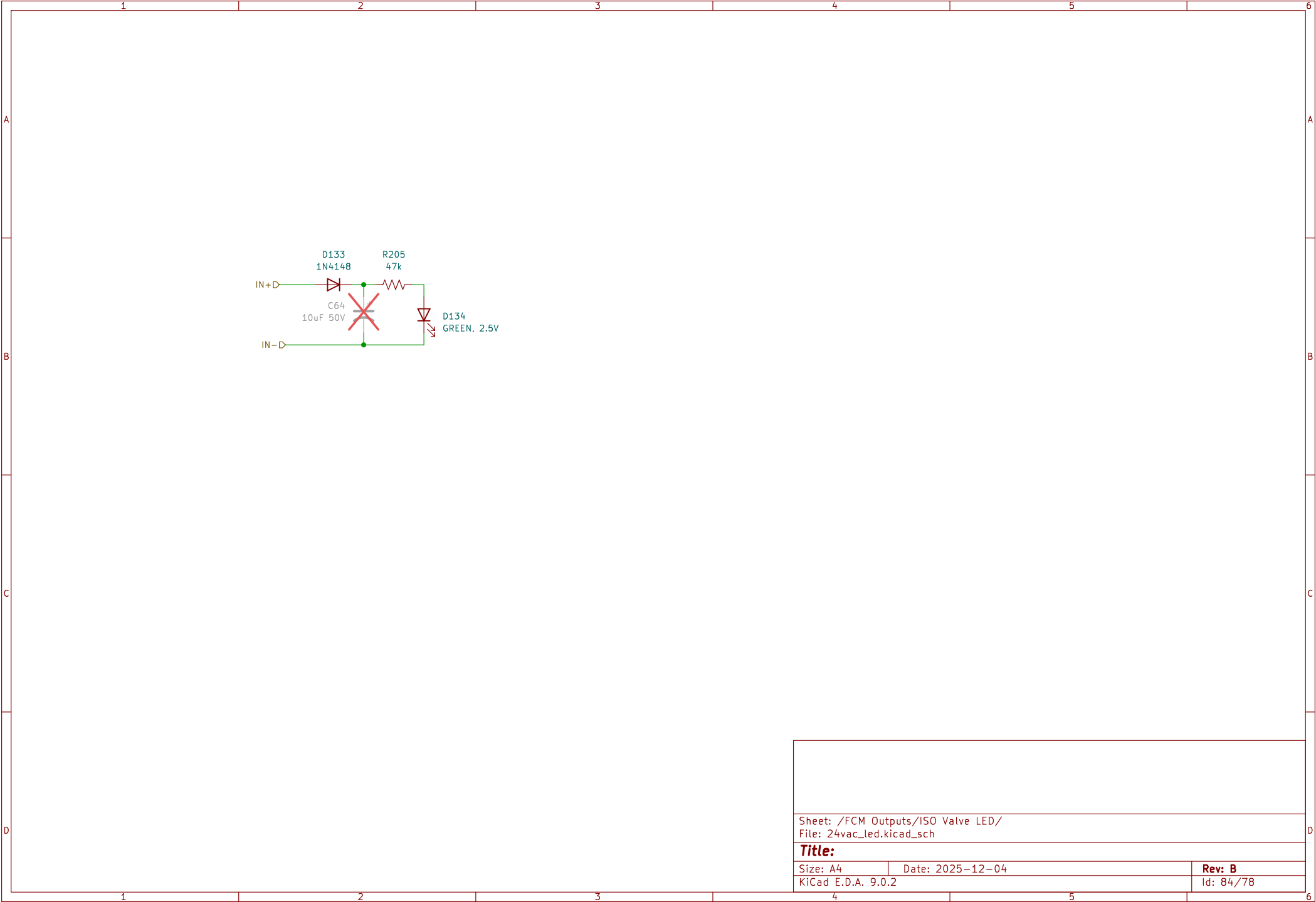
KiCad E.D.A. 9.0.2

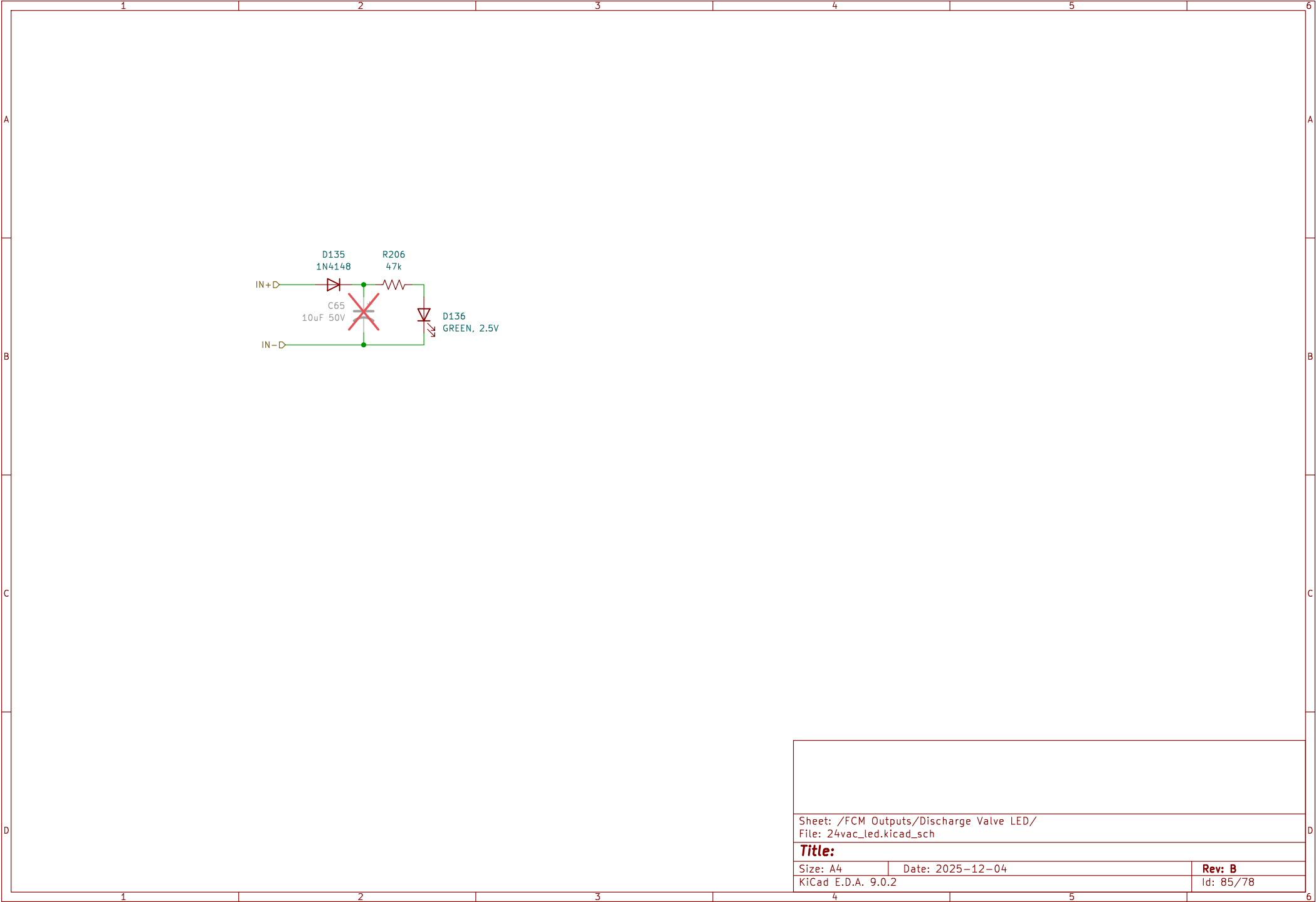
Id: 80/78

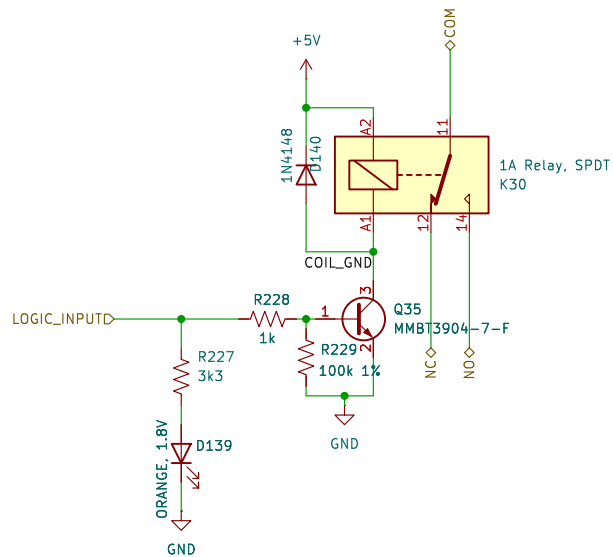












Sheet: /FCM Outputs/FCM Misc Relay/
File: signal_relay.kicad_sch

Title:

Size: A4

Date: 2025-12-04

Rev: B

KiCad E.D.A. 9.0.2

Id: 86/78